Wave State Propagation for Irregular-Time Neural Dynamics Modeling

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Abstract

Sequence learning from irregular observations requires models to distinguish elapsed physical time from observation order. We introduce Wave State Propagation (WSP), a wave-parameterized neural approach combining content-conditioned injection with persistent evolution over measured intervals. Derived from Wave Function Backpropagation, WSP uses amplitude A, latent wavenumber k, and phase θ to construct wave states, while angular frequency ω and damping $\exp(-\gamma\Delta t/T)$ govern closed-form propagation at a training-derived time scale T. An exact paired-real implementation retains both quadratures, and a zero-time displacement readout connects latent evolution to prediction. Evaluation on four physical-dynamics tasks, a physiological prediction task, and a time-irrelevant control combines parameter-matched neural comparisons with component ablations using ten final training seeds per setting. Persistent WSP reduces mean test normalized root-mean-square error versus a time-conditioned Transformer by 35.1% on HalfCheetah, 10.6% on Reacher, and 7.2% on short-window Hopper. Closely related diagonal state-space models remain competitive, and simpler WSP variants improve some tasks. Shuffled timing degrades WSP on time-relevant tasks but has negligible effect on the control. With validation error matched within 1% on PhysioNet, batch-256 WSP inference uses 50.9% less peak allocated memory than Transformer, whose throughput is 22.4 times higher on an NVIDIA B200. These findings support WSP as a viable approach to sequence prediction involving continuous physical time, with application-dependent accuracy

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and memory–throughput trade-offs.

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1. Introduction

Sequence order and physical time are not interchangeable. Positional encoding indicates whether one observation precedes another, whereas an elapsed interval specifies how long the underlying system evolves. Two records can therefore have identical content and order but imply different transitions when their sampling intervals differ. This distinction is central to asynchronous sensing, event streams, robotics, and continuous control. A model that treats Δt only as another feature must learn the corresponding flow map indirectly.

Continuous-time neural models address this problem in several ways. Time-dependent decay modifies recurrent states according to observation gaps [1]; latent ordinary differential equations and neural controlled differential equations evolve hidden variables between observations [2, 3]; and ContiFormer combines continuous trajectories with attention [4]. Structured state-space models (SSMs) offer a complementary closed-form route: their diagonal or diagonalizable continuous generators support stable recurrence and parallel scans [5–8]. Selective SSMs further make transition parameters input-dependent [9]. These precedents make the relationship between WSP and complex diagonal SSMs important: WSP's homogeneous physical-time flow belongs to that mathematical family. Its proposed distinction is the WFB-derived content phase and amplitude, the use of measured physical intervals, and a zero-time displacement construction, not the claim that diagonal complex recurrence is itself new.

Wave Function Backpropagation (WFB) introduced a wave response with learnable amplitude A, wavenumber $\mathbf{k}$, angular frequency ω, and phase θ [10]. For input $\mathbf{x}$ and elapsed time Δt, WFB uses the phase $\mathbf{k}^\top \mathbf{x} - \omega \Delta t - \theta$. Its feed-forward evaluation recomputes that response at each sample; it does not carry a wave state through an ordered observation sequence.

This paper introduces *Wave State Propagation* (WSP). The evaluated physical-time operator is $\exp(-\gamma \Delta t/T)\exp(-\mathrm{i}\omega \Delta t/T)$, where T is a training-derived time scale. Wavenumber k and phase θ determine the content-dependent injection phase, whereas ω and γ determine rotation and atten-

uation over measured time. The current observation initializes or updates the wave state, which is then propagated over its associated interval. Paired real-valued cosine and sine channels implement the complex recurrence exactly. We study both a transition-local specialization, which initializes a wave for each supervised transition, and a persistent specialization, which carries state through ordered windows. Keeping these variants separate is necessary because individual-transition prediction does not by itself test across-observation persistence.

The primary empirical study evaluates persistent WSP on a damped oscillator, HalfCheetah, Reacher, short-window Hopper, a PhysioNet sequence-prediction adaptation, and a synthetic control whose target does not depend on elapsed time. Validation-only hyperparameter selection precedes ten-seed test comparisons with closely related diagonal SSMs, temporal recurrent models, and a time-conditioned, zero-time-constrained Transformer. Component ablations examine persistence, damping, rotation, quadratures, conditioning, and the zero-time readout. A separate single-transition protocol tests the local specialization. Supplemental comparisons with f-CRU, LRU, Phased LSTM, and UnICORNN extend the range of temporal baselines under a separately declared tuning protocol. These experiments are not pooled into additional independent seed replications.

The contributions of this work are threefold:

1. We formulate a WFB-derived neural sequence model that separates content-dependent wave injection from evolution over measured physical time, with transition-local and persistent specializations.
2. We give its exact paired-real implementation and zero-time displacement readout, and establish the fixed-generator identity, homogeneous stability, and affine-composition properties within the diagonal neural SSM family.
3. We quantify predictive accuracy, temporal sensitivity, open-loop rollout, and accuracy-matched inference costs under a declared parameter and optimization budget. The comparisons identify when persistent wave dynamics improve prediction and when simpler temporal models are competitive, including the workload-dependent trade-off between inference memory and throughput.

The intended contribution is a concrete, differentiable parameterization of time-aware sequence dynamics, not a universal replacement for existing temporal models. Its usefulness is determined by the application: observed

accuracy gains, the value of persistent state and wave components, and the required inference operating point.

The remainder of the paper is organized as follows. Section 2 reviews continuous-time sequence models, neural SSMs, and world models. Section 3 summarizes the wave-function foundation inherited from WFB. Section 4 develops WSP and its mathematical properties. Section 5 presents the experimental protocol and results. Section 6 concludes the paper and identifies directions for extending WSP to deeper sequence and control architectures.

2. Related Work

2.1. Irregular and Continuous-Time Sequence Learning

Conventional recurrent and attention-based models operate primarily over an ordinal sequence index. Irregularly sampled data require an additional model of the duration between observations. GRU-D introduces interval-dependent decay to handle missing multivariate measurements [1]. Latent ODEs instead infer a latent initial state and evolve it with a neural ordinary differential equation [2, 11], while GRU-ODE-Bayes and liquid time-constant networks combine recurrent updates with continuous flows [12, 13]. Phased LSTM gates updates with a learned oscillatory clock for sparse event streams [14]. Continuous Recurrent Units (CRUs) use continuous-discrete Kalman updates and closed-form latent dynamics; the efficient f-CRU variant restricts the transition parameterization [15]. UnICORNN derives a recurrent architecture from a structure-preserving discretization of coupled oscillator dynamics [16]. TGNN4I combines recurrent observation updates with closed-form decay and periodic dynamics for irregular temporal graphs [17]. These methods establish the continuous-time and oscillator foundations on which WSP builds. Neural controlled differential equations use an interpolated observation path to drive continuous hidden dynamics [3]. These models represent important continuous-time alternatives to fixed-step recurrence. WSP combines a content-dependent amplitude and phase injection with closed-form propagation whose angular frequency and damping determine how each measured interval transforms the hidden state. The evaluation includes a preprocessed PhysioNet sequence-prediction task, but does not assess clinical outcomes, observation-mask-aware imputation, or calibrated stochastic latent inference.

2.2. Structured State-Space Models

S4, diagonal state-space layers, and S5 show that continuous linear generators can be discretized into efficient long-sequence recurrences [5–7]. LRU obtains related benefits with a carefully parameterized diagonal complex recurrence, and Mamba makes SSM parameters input-selective [8, 9]. For a fixed interval and conditioning context, WSP's homogeneous multiplier is a diagonal complex SSM. WSP differs in how WFB variables initialize the content phase, how measured Δt directly discretizes the flow, and how the model reads the change from its zero-time wave state. The primary benchmark directly tests global and content-conditioned diagonal-SSM controls with the same propagation and readout structure. A separately configured supplemental study includes a parameter-matched LRU adaptation. S5 supplies related mathematical context rather than a native benchmark implementation. Related Koopman approaches also seek latent coordinates with simple linear evolution, but generally learn the embedding rather than prescribe a WFB phase parameterization [18].

2.3. Continuous-Time Attention and World Models

Transformers use attention to model long-range dependencies [19]. Positional encoding alone does not specify elapsed physical time. ContiFormer extends attention to irregular continuous-time data by combining neural ODE trajectories with attention integration [4]. WSP shares the objective of modeling continuous evolution but replaces numerical vector-field integration with an analytic, state-conditioned phase propagation.

Learned world models provide another route to continuous-control dynamics. TD-MPC2 learns a decoder-free latent model used for local trajectory optimization [20]. DCWM introduces stochastic discrete codebook states and combines them with decision-time planning in DC-MPC [21]. Their native objectives include latent consistency, reward or value prediction, policies, and planning. Their control objectives differ from the supervised displacement prediction studied here. Our empirical claims concern sequence prediction and recorded-action open-loop replay, not the native closed-loop return of DC-MPC or TD-MPC2.

2.4. Periodic Neural Representations and WFB

Periodic activations provide a compact basis for continuously varying and high-frequency signals. Sinusoidal representation networks, for example, show that periodic nonlinearities can encode detailed signals and their

derivatives [22]. WFB extends this perspective by jointly learning amplitude, state-dependent wavenumber, temporal angular frequency, and phase, with Δt appearing directly in the wave phase term [10]. WSP contributes two physical-time uses of this basis: a zero-time wave increment for individual transitions and a compositional carry recurrence for ordered sequences.

3. Mathematical Formulation of the Wave Function

WSP is an extension of Wave Function Backpropagation (WFB) [10], rather than an unrelated continuous-time architecture. For self-contained notation, Eqs. (1) and (2) restate the complex WFB wave response and its practical real-valued activation, respectively. They correspond to Eqs. (2) and (5), respectively, in the cited WFB paper, with channel index m made explicit here:

$$\psi_m(\mathbf{x}, \Delta t) = A_m \exp\Big[\mathrm{i}\left(\mathbf{k}_m^\top \mathbf{x} - \omega_m \Delta t - \theta_m\right)\Big], \tag{1}$$

whose real-valued activation is

$$f_m(\mathbf{x}, \Delta t) = A_m \cos\Big(\mathbf{k}_m^\top \mathbf{x} - \omega_m \Delta t - \theta_m\Big). \tag{2}$$

Equation (2) takes the real part only because the original pointwise WFB activation must return a real feature. WSP does not discard the imaginary component during propagation: it retains both quadratures and implements them as paired real tensors. The evaluated WSP code uses $+\theta$ in its injection phase; this is the harmless reparameterization $\theta_{\mathrm{WSP}} = -\theta_{\mathrm{WFB}}$ of the WFB convention above. Amplitude A_m controls response magnitude, wavenumber $\mathbf{k}_m$ controls variation with the state or spatial input, angular frequency ω_m controls phase change per unit elapsed time, and θ_m is a phase offset. WFB derives the task-loss gradients of these parameters explicitly. For compactness, their ordinary gradient-descent update can be summarized as

$$q \leftarrow q - \eta \frac{\partial \mathcal{L}_{\mathrm{task}}}{\partial q}, \qquad q \in \{A_m, \mathbf{k}_m, \omega_m, \theta_m\}. \tag{3}$$

Equation (3) is a standard optimization summary, not an additional wave equation quoted verbatim from WFB. The WFB proof of concept applies Eq. (1) as an interval-conditioned representation inside an FFN. It therefore changes the activation at each observation but does not preserve and propagate a wave state between successive observations.

4. Wave State Propagation

This section gives the formulation used by the reported implementations. WSP contains two specializations: a transition-local wave increment for individual-transition prediction and a persistent carry recurrence used by the primary ordered-window benchmark. Both use the same physical-time operator, but only the second carries a state across observations.

4.1. Physical-Time Generator

For transition i, observation $\mathbf{x}_i$ is recorded at time τ_i and evolves to $\mathbf{x}_{i+1}$ over

$$\Delta t_i = \tau_{i+1} - \tau_i, \qquad \widetilde{\Delta t}_i = \frac{\Delta t_i}{T}, \tag{4}$$

where T is the largest training interval. Ordinal position may be encoded by a surrounding sequence model, but the evaluated WSP dynamics has no independent positional propagation coordinate.

For wave channel m and layer ℓ, the homogeneous continuous dynamics over one interval are expressed in normalized time $u = t/T$:

$$\frac{\mathrm{d}\psi_m^{(\ell)}}{\mathrm{d}u} = \left(-\gamma_{i,m}^{(\ell)} - \mathrm{i}\omega_{i,m}^{(\ell)}\right) \psi_m^{(\ell)}, \tag{5}$$

with the state-conditioned parameters held fixed during that interval. Equivalently, the physical-time generator is $(-\gamma_{i,m}^{(\ell)} - \mathrm{i}\omega_{i,m}^{(\ell)})/T$. The exact flow for normalized duration $u \geq 0$ is

$$\mathcal{P}_i^{(\ell)}(u) = \mathrm{diag}\Big[\exp\Big(-\boldsymbol{\gamma}_i^{(\ell)} u\Big) \odot \exp\Big(-\mathrm{i}\boldsymbol{\omega}_i^{(\ell)} u\Big)\Big]. \tag{6}$$

Thus ω controls physical-time rotation and γ controls attenuation. The inherited wavenumber k does not appear in Eq. (6); in the evaluated models it controls the latent content-injection phase. This distinction corrects a possible but inaccurate interpretation of k as an additional token-position transition.

Equation (5) also makes the relationship to neural SSMs explicit. Its generator is a diagonal complex state matrix, so the homogeneous WSP flow is mathematically a complex diagonal continuous-time SSM [6–8]. WSP's contribution lies in its WFB-derived content parameterization, measured-time discretization, zero-time displacement readout, and persistent injection rule.

4.2. State-Conditioned Wave Parameters

Let $\mathbf{h}_i^{(\ell-1)} \in \mathbb{R}^{d_h}$ be the hidden content and $\mathbf{c}_i^{(\ell)} = \mathrm{LN}(\mathbf{h}_i^{(\ell-1)})$. A rank-16 conditioner produces five channel-wise shifts,

$$[\boldsymbol{\delta}_A, \boldsymbol{\delta}_k, \boldsymbol{\delta}_\omega, \boldsymbol{\delta}_\gamma, \boldsymbol{\delta}_\theta]_i^{(\ell)} = \mathrm{split}_5\Big(\mathbf{W}_{\mathrm{up}}^{(\ell)} \mathbf{W}_{\mathrm{down}}^{(\ell)} \mathbf{c}_i^{(\ell)}\Big). \tag{7}$$

The evaluated parameterization is

$$\begin{aligned} \mathbf{A}_i^{(\ell)} &= \mathrm{softplus}(\mathbf{a}^{(\ell)} + \boldsymbol{\delta}_A), & \mathbf{k}_i^{(\ell)} &= \mathrm{softplus}(\boldsymbol{\kappa}^{(\ell)} + \boldsymbol{\delta}_k), \\ \boldsymbol{\omega}_i^{(\ell)} &= \omega_{\max}\sigma(\mathbf{o}^{(\ell)} + \boldsymbol{\delta}_\omega), & \boldsymbol{\gamma}_i^{(\ell)} &= \gamma_{\max}\sigma(\mathbf{r}^{(\ell)} + \boldsymbol{\delta}_\gamma), \\ \boldsymbol{\theta}_i^{(\ell)} &= \boldsymbol{\theta}_0^{(\ell)} + \pi \tanh(\boldsymbol{\delta}_\theta), \end{aligned} \tag{8}$$

where $\omega_{\max} = 2\pi$ and $\gamma_{\max} = 2$ in normalized time. Amplitude A, latent wavenumber k, angular frequency ω, damping γ, and phase θ are therefore recomputed from the current content at each layer. They are not recomputed within the interval governed by Eq. (6).

4.3. Transition-Local Zero-Time Wave Increment

The transition-local specialization initializes a wave from each state–action pair and does not carry that wave to the next training sample. At layer ℓ,

$$\boldsymbol{\psi}_{i,0}^{(\ell)} = \mathbf{A}_i^{(\ell)} \odot \mathbf{c}_i^{(\ell)} \odot \exp\Big\{\mathrm{i}\left[\mathbf{k}_i^{(\ell)} \odot \mathbf{c}_i^{(\ell)} + \boldsymbol{\theta}_i^{(\ell)}\right]\Big\}. \tag{9}$$

For L stacked layers, each layer advances the wave by $u_i = \widetilde{\Delta t}_i / L$ and reads the change from its zero-time state:

$$\begin{aligned} \boldsymbol{\delta\psi}_i^{(\ell)} &= \left[\mathcal{P}_i^{(\ell)}(u_i) - \mathbf{I}\right] \boldsymbol{\psi}_{i,0}^{(\ell)}, \\ \mathbf{h}_i^{(\ell)} &= \mathbf{h}_i^{(\ell-1)} + \mathbf{W}_r^{(\ell)} \,\mathrm{vec}_{\mathbb{R}}\Big(\boldsymbol{\delta\psi}_i^{(\ell)}\Big), \end{aligned} \tag{10}$$

where $\mathrm{vec}_{\mathbb{R}}(\psi)$ stores both real and imaginary quadratures in a fixed ordering matched to the readout weights. With $\mathbf{h}_i^{(0)} = \mathbf{E}[\mathbf{s}_i; \mathbf{a}_i] + \mathbf{b}_E$, the normalized displacement prediction is

$$\widehat{\mathbf{y}}_i = \mathbf{W}_o\left[\mathrm{LN}(\mathbf{h}_i^{(L)}) - \mathrm{LN}(\mathbf{h}_i^{(0)})\right] - \boldsymbol{\mu}_{\Delta s} \oslash \boldsymbol{\sigma}_{\Delta s}. \tag{11}$$

At $\Delta t_i = 0$, every wave increment is zero. Decoding Eq. (11) with the training target statistics therefore gives exactly zero displacement in raw state units. This specialization evaluates the elapsed-time wave operator

without across-observation carry. The primary sequence and replay results instead use the persistent model below. The single-transition test results are reported separately, so the performance of one specialization is not attributed to the other.

The evaluated head in Eq. (11) is bias-free and linear, without a separate ReLU/GELU multilayer decoder. This is an architectural choice for the displacement task, not a claim that every application of WSP can omit a task-specific decoder.

4.4. Persistent Injection-Then-Propagation Recurrence

The ordered-window study evaluates a distinct persistent recurrence. From the normalized content, a learned drive and gate form

$$\begin{aligned} \mathbf{d}_i^{(\ell)} &= \mathbf{A}_i^{(\ell)} \odot \left(\mathbf{D}^{(\ell)}\mathbf{c}_i^{(\ell)} + \mathbf{b}_D^{(\ell)}\right), & \mathbf{g}_i^{(\ell)} &= \sigma\left(\mathbf{G}^{(\ell)}\mathbf{c}_i^{(\ell)} + \mathbf{b}_g^{(\ell)}\right), \\ \mathbf{s}_i^{(\ell)} &= \mathbf{d}_i^{(\ell)} \odot \exp\left\{\mathrm{i}\left[\mathbf{k}_i^{(\ell)} \odot \mathbf{c}_i^{(\ell)} + \boldsymbol{\theta}_i^{(\ell)}\right]\right\}. \end{aligned} \tag{12}$$

The current content is injected before physical-time propagation:

$$\begin{aligned} \overline{\boldsymbol{\psi}}_i^{(\ell)} &= \boldsymbol{\psi}_{i-1,+}^{(\ell)} + \mathbf{g}_i^{(\ell)} \odot \mathbf{s}_i^{(\ell)}, \\ \boldsymbol{\psi}_{i,+}^{(\ell)} &= \mathcal{P}_i^{(\ell)}(\widetilde{\Delta t}_i)\overline{\boldsymbol{\psi}}_i^{(\ell)}, \\ \mathbf{h}_i^{(\ell)} &= \mathbf{h}_i^{(\ell-1)} + \mathbf{W}_r^{(\ell)} \operatorname{vec}_{\mathbb{R}}\left(\boldsymbol{\psi}_{i,+}^{(\ell)} - \overline{\boldsymbol{\psi}}_i^{(\ell)}\right). \end{aligned} \tag{13}$$

Here "+" denotes the state after propagation over transition i, not a positive-frequency component. The sequence study uses two layers, each with the full normalized interval. Its reset ablation sets the incoming carry to zero at every transition; its carry model passes $\boldsymbol{\psi}_{i,+}^{(\ell)}$ to the next observation. The persistent model uses the readout in Eq. (11) with its recurrent $\mathbf{h}_i^{(L)}$, preserving zero raw displacement at zero interval. Carry is initialized to zero at the start of each evaluation window; the benchmark does not assume persistence between windows.

4.5. Exact Paired-Real Implementation

Complex arithmetic is not required. For $\psi = u + \mathrm{i}v$, define $\mathbf{z} = [u, v]^\top$ and

$$\mathbf{R}(\phi) = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}. \tag{14}$$

Here $u_i = \widetilde{\Delta}t_i/L$ for the transition-local variant and $u_i = \widetilde{\Delta}t_i$ for the persistent variant. Then one WSP channel is implemented exactly as

$$\mathbf{z}^{(\ell)}_{i,+,m} = \exp\!\big(-\gamma^{(\ell)}_{i,m} u_i\big)\, \mathbf{R}\big(-\omega^{(\ell)}_{i,m} u_i\big)\, \bar{\mathbf{z}}^{(\ell)}_{i,m}. \tag{15}$$

Thus the full WSP uses real tensors but preserves both quadratures. The "cosine-only" ablation discards the sine quadrature and is not the exact real representation of the complex operator.

4.6. Properties and Parallel Scan

For a fixed conditioned generator,

$$\mathcal{P}(0) = \mathbf{I}, \qquad \mathcal{P}(u_2)\mathcal{P}(u_1) = \mathcal{P}(u_1 + u_2). \tag{16}$$

When the conditioner changes between observations, the second identity does not imply one global semigroup; it applies only while the generator is held fixed. For nonnegative damping and $u \geq 0$,

$$\|\mathcal{P}(u)\|_2 \leq 1. \tag{17}$$

Equality is possible for an undamped channel. This is a bound on the homogeneous flow, not on arbitrary driven input. For the affine recurrence $\boldsymbol{\psi}_i = \mathbf{P}_i \boldsymbol{\psi}_{i-1} + \mathbf{b}_i$ with $\|\mathbf{P}_i\|_2 \leq 1$, a simple driven bound is $\|\boldsymbol{\psi}_n\|_2 \leq \|\boldsymbol{\psi}_0\|_2 + \sum_{i=1}^{n} \|\mathbf{b}_i\|_2$.

With conditioner outputs held fixed, the direct derivatives of one evaluated operator entry are

$$\frac{\partial [\mathcal{P}_i]_{mm}}{\partial \omega_{i,m}} = -\mathrm{i} u_i [\mathcal{P}_i]_{mm}, \qquad \frac{\partial [\mathcal{P}_i]_{mm}}{\partial \gamma_{i,m}} = -u_i [\mathcal{P}_i]_{mm}. \tag{18}$$

Wavenumber k receives gradients through Eq. (9) or Eq. (12), not through the evaluated physical-time operator.

For the persistent injection-before-propagation rule, define $\mathbf{b}_i := \mathbf{P}_i(\mathbf{g}_i \odot \mathbf{s}_i)$. Two affine steps compose as

$$(\mathbf{P}_2, \mathbf{b}_2) \circ (\mathbf{P}_1, \mathbf{b}_1) = (\mathbf{P}_2\mathbf{P}_1, \mathbf{P}_2\mathbf{b}_1 + \mathbf{b}_2). \tag{19}$$

This operation is associative, so a layer can in principle use a parallel prefix scan. The reported persistent implementation is serial; the scan is a mathematical implementation path and is not used to support the measured throughput claims.

All reported WSP models are trained end-to-end with the normalized displacement loss and ordinary backpropagation. No wave-equation residual, dispersion penalty, energy penalty, or Laplacian parameter update is used. Transformer fusion and phase-aware attention are possible extensions, but they are outside the evaluated architecture and are therefore not included in the claimed method.

5. Evaluation

5.1. Questions, Tasks, and Data

The evaluation tests the value of explicitly propagating a learned state over measured physical intervals. We examine prediction, component effects, elapsed-time sensitivity, autoregressive replay, and the accuracy–throughput–memory trade-off. The primary study contains 1,460 completed final evaluations: 850 ordered-sequence evaluations on five controlled dynamics datasets, 500 transition-local evaluations on those datasets, and 110 sequence evaluations on PhysioNet. Each model–dataset configuration uses ten final seeds. The two WSP specializations have distinct information sets and are analyzed separately.

The controlled datasets are a damped oscillator, Reacher-v5, HalfCheetah-v5, Hopper-v5, and a Δt-irrelevant control. MuJoCo tasks [23] hold an action over a known number of physics steps, giving $\Delta t_i = n_i h$. Counterfactual branches share an anchor state and action but vary the interval; only the behavior branch advances the episode and enters an ordered sequence. Episode seeds are disjoint across training, validation, and test. The negative control uses a nonlinear discrete state–action transition independent of the supplied interval. It detects unnecessary timing reliance, rather than requiring a time-aware model to perform poorly.

The PhysioNet adaptation uses irregular records from Set A of the PhysioNet/Computing in Cardiology Challenge 2012 [24, 25]. Ten physiological channels are aligned at the union of their observation times. Previous measurements are carried forward, with training-patient medians initializing not-yet-observed channels. Patients are partitioned disjointly, time is expressed in seconds, and the action channel is zero. All models receive the same prepared values without observation masks. This is dense displacement prediction, not mortality prediction, mask-aware imputation, or evidence of clinical utility.

An ordered sample provides $(\mathbf{s}_i, \mathbf{a}_i, \Delta t_i)$ and predicts $\mathbf{s}_{i+1} - \mathbf{s}_i$. The upcoming interval is a known forecast input; the task does not predict the next event time. WSP injects current content before propagating over that interval, as in Eq. (13), rather than interpreting it as the gap preceding the current state. Sequence windows contain 32 transitions except Hopper, whose shorter episodes use length 4. Windows never cross episodes, incomplete tails are excluded, and carry is initialized at each window. Transition-local models score individual transitions without across-observation carry.

State, action, and displacement normalization uses training statistics only. For coordinate j, the target and error metric are

$$y_{i,j} = \frac{\Delta s_{i,j} - \mu_{\Delta s,j}}{\sigma_{\Delta s,j}}, \qquad \Delta \mathbf{s}_i = \mathbf{s}_{i+1} - \mathbf{s}_i, \tag{20}$$

$$\mathrm{NRMSE}_{\Delta s} = \sqrt{\frac{1}{ND} \sum_{i=1}^{N} \sum_{j=1}^{D} (\widehat{y}_{i,j} - y_{i,j})^2}. \tag{21}$$

NRMSE is dimensionless standardized-displacement error, not an accuracy percentage. Test distributions include intervals beyond the training range; values above one must therefore be interpreted against explicit simple baselines, not an assumed universal threshold.

5.2. Comparators, Selection, and Uncertainty

The primary controls are global and content-conditioned diagonal SSMs, time-GRU, a TGNN4I-inspired probe, time-conditioned Transformers, and simple predictors. The diagonal SSMs retain paired quadratures, damped rotation, and the zero-time output construction, but use affine injection. They directly contextualize WSP's content-dependent amplitude, phase, and gated drive, without constituting native S5 or LRU reproductions. The TGNN4I-inspired probe combines a GRU observation update with channel-wise continuous dynamics; it does not reproduce the native graph architecture [17]. Persistence, training-mean displacement, ridge, MLP, and available constant-velocity controls expose cases where a simpler predictor succeeds.

The sequence Transformer uses causal temporal attention, positional and time features, and a readout constrained to zero raw displacement at zero current interval. The term *zero-time* denotes this boundary, not omission of timing information. Transition experiments distinguish this architecture applied to a single transition from a feature-token Transformer attending

across state/action feature tokens. Neither transition comparator observes a temporal sequence.

Persistent WSP uses width 128, two layers, and a rank-16 conditioner; transition-local WSP uses four layers. Neural comparator widths target the corresponding WSP parameter budget within 5%. Achieved discrepancies are at most 1.21% for controlled sequences, 2.24% for transitions, and 0.915% for PhysioNet. Fixed predictors are not forced to match this budget (Appendix Table A.1).

The 1,386 development evaluations search learning rates $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$ and weight decays $\{0, 10^{-4}\}$ with seeds 1–3. Selection minimizes mean validation NRMSE without test evaluation. Development uses at most 50 epochs and patience 10; final seeds 101–110 use at most 100 epochs and patience 20. WSP ablations inherit the full model's selected settings. AdamW, cosine learning-rate decay, clipping at 1, and batch size 256 are shared. Independent jobs use two B200 GPUs, PyTorch 2.9.1/CUDA 13.0, unfused AdamW, and a mixed-precision BF16 policy. Matrix-multiplication TF32 is disabled; cuDNN TF32 is enabled. Numerical runtime checks precede training. Selected settings are in Appendix Table A.10.

We aggregate elements within each seed, then report seed means and sample standard deviations. Paired bootstrap intervals use 20,000 resamples; two-sided sign-flip tests are Holm-adjusted within the recorded families of 80 controlled-sequence, 45 transition, and 10 PhysioNet contrasts. None of the first two families passes adjusted $p < 0.05$; seven PhysioNet contrasts do. With ten pairs, the smallest exact two-sided sign-flip probability is $2/2^{10}$, limiting resolution after adjustment over large families. Main-task effects and unadjusted intervals are consequently described without claiming corrected significance. Uncertainty concerns training seeds on fixed partitions, not new datasets; repeated deterministic predictors add no independent training variability. Appendix Tables A.4–A.6 retain all tests.

5.3. Predictive Accuracy and Closest Controls

Among the physical-dynamics tasks, HalfCheetah gives WSP's largest mean reduction relative to attention: NRMSE is 0.71304 versus 1.09851 for Transformer, a 35.1% reduction. Mean reductions on Reacher and short-window Hopper are 10.6% and 7.2%, respectively. The conditioned-SSM contrast is narrower: HalfCheetah reaches 0.73071 versus WSP's 0.71304, a 2.4% mean reduction, with paired SSM-minus-WSP interval $[-0.00107, 0.04398]$. WSP and conditioned SSM are also close on Reacher (1.24418 versus 1.24604)

and Hopper (3.15518 versus 3.19461). Thus a favorable attention comparison does not isolate a large advantage over matched damped-rotation dynamics.

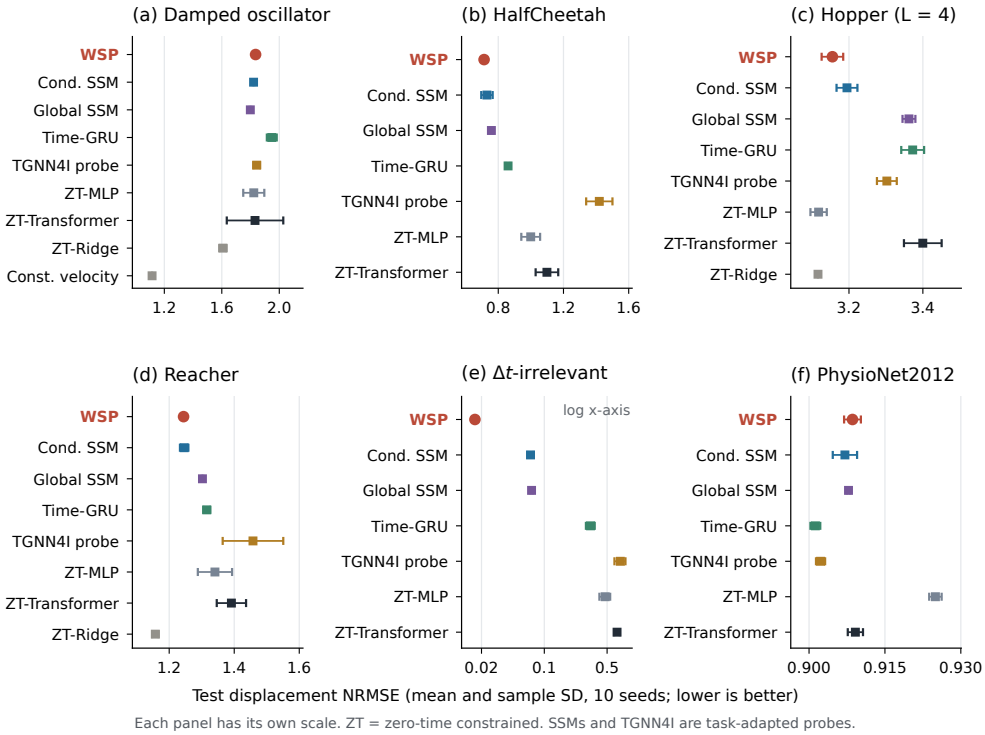


Figure 1: Primary sequence prediction across six datasets. Points and bars show mean test displacement NRMSE and sample standard deviation over ten final seeds. Lower is better; panel scales are separate. Simple predictors and close diagonal-SSM controls contextualize neural comparisons. Hopper uses length 4; other datasets use length 32. Paired inference is reported separately.

The oscillator supplies a counterexample: WSP (1.83519) is close to Transformer (1.83125), while constant velocity achieves 1.11475. Ridge also improves on WSP on Reacher (1.15743 versus 1.24418). On the time-irrelevant control, WSP reaches 0.01690, versus 0.07025 for conditioned SSM and 0.64546 for Transformer. Explicit-time parameterization thus need not prevent learning a largely time-independent map; this control does not demonstrate a benefit caused by informative time.

PhysioNet WSP reaches 0.90859 ± 0.00166, close to Transformer (0.90916 ± 0.00151; adjusted $p = 0.4121$). Time-GRU (0.90125) and

TGNN4I probe (0.90228) are better on all ten seeds, each with adjusted $p = 0.0195$. WSP is competitive but not leading on this prepared real-data task.

Transition-local WSP has the lowest mean oscillator error among its comparators (1.40445), but not elsewhere: conditioned SSM wins Reacher, ridge wins HalfCheetah, feature-token Transformer wins Hopper, and MLP wins the time-irrelevant control. HalfCheetah transition WSP (1.24922) exceeds the training-mean error (1.17865). Sequence and transition scores are not interchangeable because sampling and available history differ. Appendix Tables A.1–A.3 report every comparator.

5.4. What the Component Controls Establish

Figure 2 uses the completed ten-seed controls. Resetting carry raises mean NRMSE from 0.71304 to 1.02731 on HalfCheetah and from 1.24418 to 1.30113 on Reacher; the effect is small on the oscillator and short-window Hopper. PhysioNet carry improves on reset (0.90859 versus 0.92575, adjusted $p = 0.0195$). Persistence is useful in several tasks, without a universal gain at every window length.

Removing damping (rotation-only) worsens all five controlled-task means. Rotation itself is not uniformly beneficial: decay-only improves HalfCheetah from 0.71304 to 0.70336 and Hopper from 3.15518 to 2.72765. Conversely, full WSP improves on decay-only in PhysioNet (0.90859 versus 0.91220, adjusted $p = 0.0234$). Removing conditioning or the zero-time constraint also improves several means: HalfCheetah reaches 0.69512 and 0.68808, respectively. These controls identify task-dependent design choices, not universal necessity of every wave component. The zero-time construction remains an exact boundary property even when it does not minimize test error.

5.5. Use of Elapsed Time

For fixed checkpoints and unchanged targets, shuffling supplied intervals increases WSP mean NRMSE by 0.10257 on the oscillator, 0.10045 on HalfCheetah, 0.08086 on Reacher, 0.18284 on Hopper, and 0.00589 on PhysioNet. Each paired bootstrap interval is positive. The time-irrelevant change is -0.000081, with interval $[-0.000241, 0.000075]$, consistent with negligible shuffled-time reliance. Figure 3 scales changes by each seed's correct-time error; Appendix Table A.7 retains absolute differences.

Other temporal models also deteriorate under shuffling, so the test verifies timing use rather than uniquely identifying WSP's mechanism. Setting

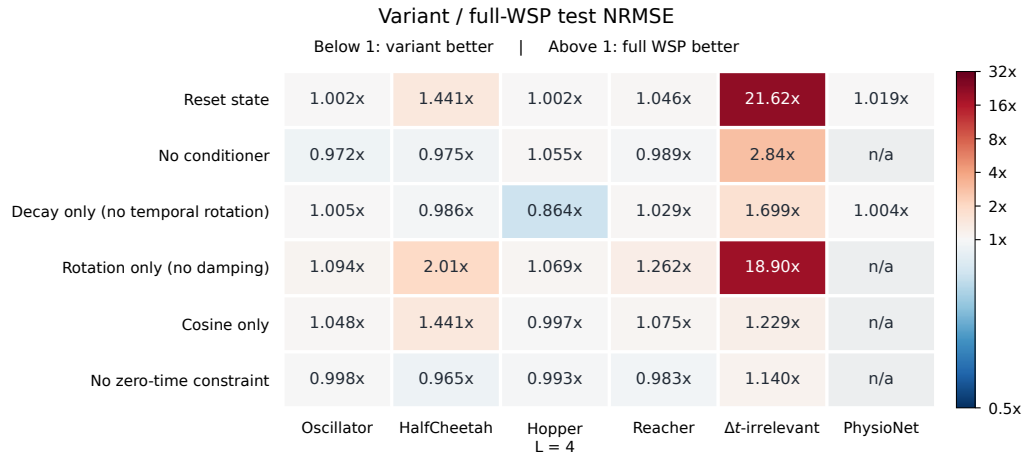


Figure 2: Component controls relative to full carry mean NRMSE. Ratios above one indicate deterioration. Comparisons use ten final seeds and the full model's selected optimization settings. Decay-only removes temporal rotation but retains content injection; removing the zero-time constraint differs from setting the time input to zero. Only reset and decay-only are evaluated for PhysioNet.

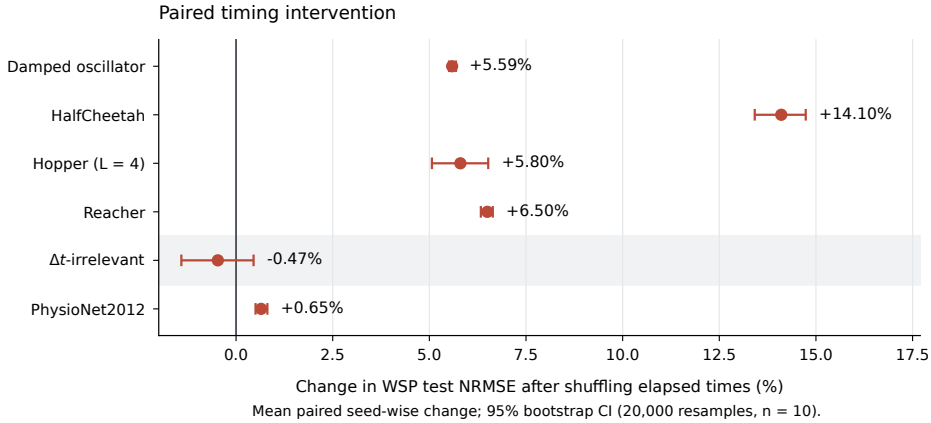


Figure 3: Relative change in WSP test NRMSE when elapsed-time inputs are shuffled while targets remain fixed. Points summarize paired seed-level percentage changes; bars show 95% bootstrap intervals. Positive values indicate timing reliance. These are intervention diagnostics, not multiplicity-adjusted tests of architectural superiority.

every interval to zero is less informative: the boundary condition forces zero displacement against unchanged nonzero targets. Nor does a favorable aggregate score establish superiority at every long gap. PhysioNet contains only one evaluated above-training-maximum gap per seed, insufficient to establish clinical long-gap extrapolation.

5.6. Replay and Optimization Behavior

Replay feeds predicted states back into the model while reusing recorded actions and intervals. Its score averages state errors through horizon H, normalized by training state standard deviations:

$$\mathrm{NRMSE}_s(H) = \sqrt{\frac{1}{N_w H D} \sum_{w,h,j} \left(\frac{\widehat{s}_{w,h,j} - s_{w,h,j}}{\sigma_{s,j}} \right)^2}. \tag{22}$$

This differs from displacement NRMSE and endpoint-only error. At 32 steps, HalfCheetah WSP (0.70425) improves on Transformer (0.99949), but conditioned SSM is better (0.64652). Time-GRU is better on Reacher and PhysioNet; Hopper's four-step comparison also favors conditioned SSM over WSP. Figure 4 and Appendix Table A.8 show that one-step gains do not guarantee the best autoregressive model. Recorded-action replay is not closed-loop policy evaluation or a long-horizon stability proof.

Learning histories help interpret HalfCheetah (Figure 5). Transformer selects very early validation checkpoints despite subsequently decreasing training loss. This train–validation divergence is compatible with overfitting or distribution mismatch, not failure to reduce training error. WSP and conditioned SSM provide closer mechanistic comparators. The shared search makes the procedure reproducible, but neither establishes equally optimized architecture families nor explains the generalization gap by itself.

5.7. Accuracy-Matched Inference Efficiency

Exclusive B200 inference measurements cover seven neural models per dataset, batches 1, 16, 256, and 4096, and 30 warmups followed by 200 synchronized passes. All 4,200 controlled-sequence, 1,400 transition, and 840 PhysioNet checkpoint–workload measurements succeed. Synthetic shape-matched inputs are resident before timing. Forward-pass throughput excludes loading, host transfer, training, and application overhead. Peak allocated memory is measured in MiB, separately from reserved memory, training peaks, and total GPU allocation.

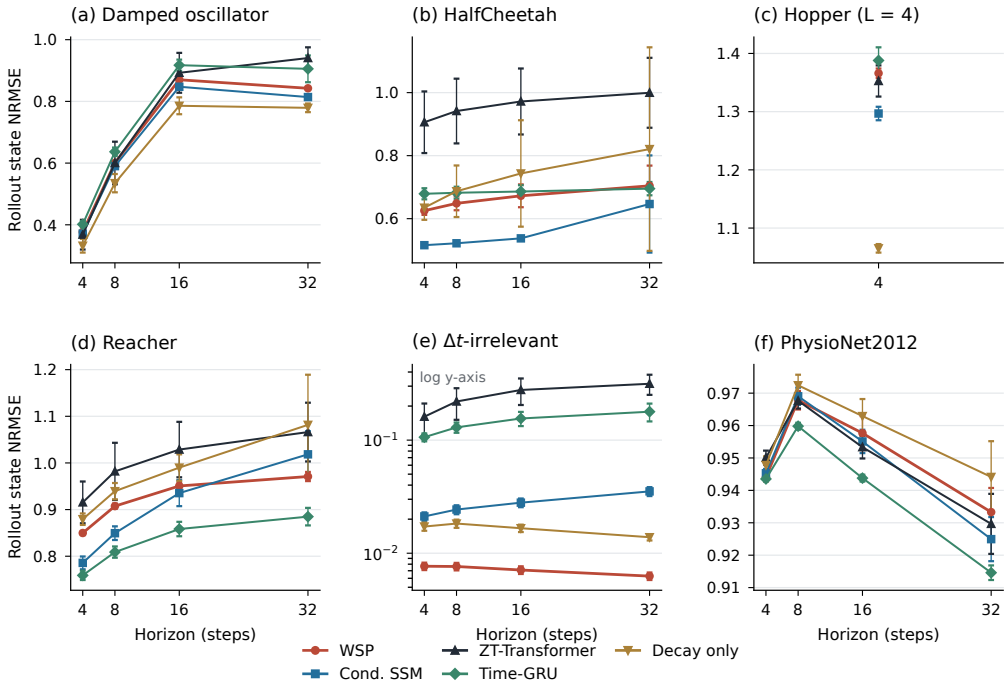


Figure 4: Open-loop state-normalized replay error across available horizons, shown as means and sample standard deviations over ten final seeds. Hopper is evaluated only through its native four-transition window. Actions are replayed, not recomputed by a controller; normalization differs from one-step displacement error.

Matching requires the same ten seeds and a relative difference between mean validation NRMSEs of at most 1%, at the native evaluation length. This is a tolerance criterion, not exact equality or a statistical equivalence test. Neither controlled-task protocol has a matched pair. PhysioNet has five matched comparators—both diagonal SSMs, TGNN4I, time-GRU, and Transformer—at four batches: 20 workload rows, not 20 independent accuracy comparisons.

At batch 256, WSP processes 222,982 transitions/s with a 14.57 MiB peak; Transformer processes 4,988,439 transitions/s with 29.69 MiB. WSP uses 50.9% less inference memory but is 22.4 times slower. At batch 4096, the memory reduction is 74.0%, with 5.86 times lower throughput. At batch 1, memory differs by only 0.24%. WSP is slower than the other matched

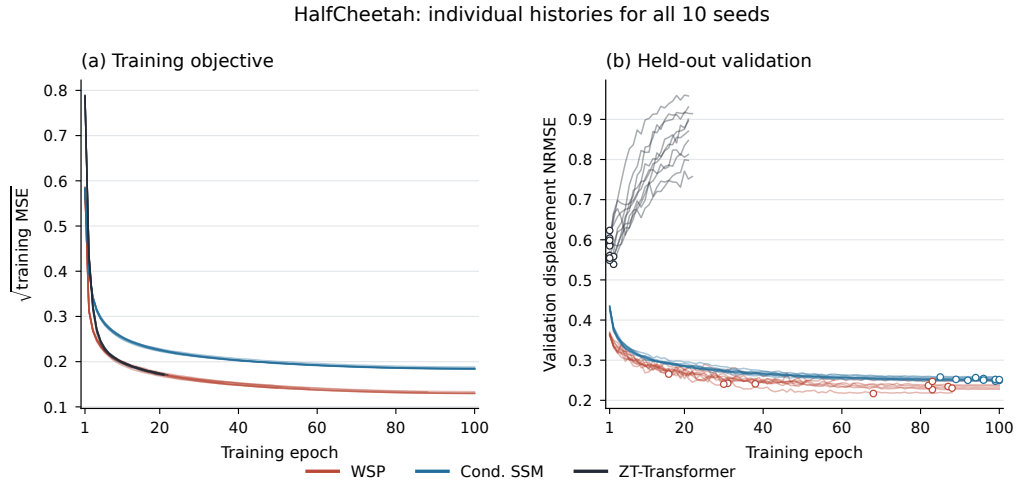


Figure 5: HalfCheetah histories for WSP, conditioned diagonal SSM, and Transformer. Individual seed traces end at their recorded stopping epochs; open circles mark selected validation checkpoints. Online training RMSE (square root of recorded training MSE) and post-epoch validation NRMSE are separate: their populations and procedures differ. Test results use validation-selected checkpoints, not last iterates.

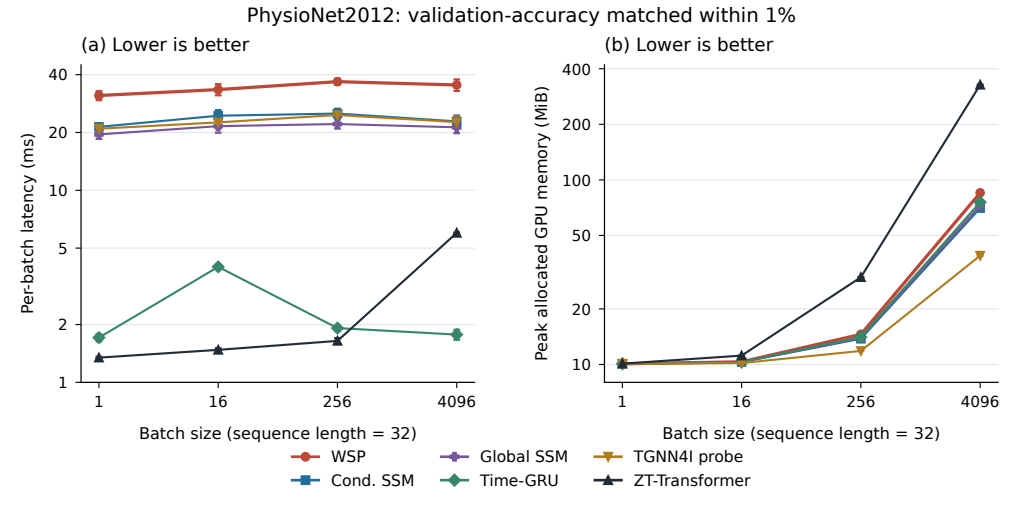


Figure 6: PhysioNet inference at native length 32 and validation NRMSE matched within 1%. Latency and peak allocated memory are shown separately across batches for the same ten final seeds. Lower values are favorable. Memory savings do not imply higher throughput or lower monetary cost.

comparators and uses more memory than those four alternatives at batch 256 (Appendix Table A.9). Its training peak is approximately 249 MiB versus 93 MiB for Transformer, reversing the inference comparison.

Transition-local WSP is 1.70–1.81 times faster than feature-token Transformer at batch 256, but uses approximately 4% more memory and is not accuracy matched. Hopper sequence inference also lacks a memory advantage at native length 4. These results establish a workload-specific trade-off, not simultaneous accuracy-matched speed and memory gains. The composition property in Eq. (19) motivates optimized implementations; it is not a measured speedup of the serial recurrence.

5.8. Separate Supplemental Recurrence Comparisons

A supplemental study compares f-CRU [15], LRU [8], Phased LSTM [14], and UnICORNN [16], with WSP and Transformer references, on the oscillator, HalfCheetah, PhysioNet, and time-irrelevant control. Its 240 final evaluations share the ten final seed identifiers but use a three-rate, fixed-weight-decay search with two development seeds and a separate implementation/runtime record. The PhysioNet preparation record also differs. Appendix Tables B.11–B.13 report this protocol separately, without pooling seeds, means, or timing.

Within that protocol, WSP has lower mean error than the four recurrent adaptations on HalfCheetah, the oscillator, and the time-irrelevant control; LRU is slightly better on PhysioNet. These are equation-based common-task adaptations, not native benchmark reproductions: f-CRU uses the shared displacement loss and FP32 covariance arithmetic, while UnICORNN lacks its authors' fused kernel. This evidence broadens recurrence coverage without replacing the primary close-SSM and component controls.

6. Conclusion and Future Work

WSP provides an explicit physical-time construction for sequence learning: WFB-derived content injection is followed by closed-form damped rotation over measured intervals, retaining both quadratures and a zero-time displacement readout. Its homogeneous dynamics belong to the complex diagonal continuous-time SSM family. The contribution is this particular content parameterization, event ordering, and readout, rather than the first use of temporal dynamics in neural sequence models.

The completed evaluation supports WSP as a viable approach for applications involving continuous physical time. Persistent WSP reduces mean prediction error relative to the evaluated time-conditioned Transformer on HalfCheetah, Reacher, and short-window Hopper; timing interventions show selective reliance on informative intervals. The closest conditioned SSM narrows these advantages, and component controls reveal when simpler variants are preferable. On PhysioNet WSP is close to Transformer but inferior to time-GRU and the TGNN4I probe. Its large-batch inference-memory saving comes with lower throughput and does not extend to training memory. These findings define a useful time-aware modeling option with application-dependent trade-offs, not a universal replacement for Transformers or recurrent state-space models.

Future work should develop numerically verified parallel kernels, test longer sequences and independently sampled datasets, extend model-specific tuning, and evaluate mask-aware real-data tasks. Closed-loop control and deployment-cost claims require dedicated experiments. These directions retain the core objective: making measured physical time an explicit, composable part of learned sequence dynamics.

Appendix A. Complete final evaluation and reproducibility

The primary study contains 850 sequence, 500 transition-local, and 110 PhysioNet final tasks. All use seeds 101–110. Tables report means and sample standard deviations across training seeds, not standard errors. Each seed first aggregates prediction elements on its fixed test partition; seed variation does not quantify uncertainty over new datasets or patient cohorts. NRMSE is dimensionless and lower is better. Displayed values are rounded; very small nonzero values use scientific notation.

Sequence windows have length 32, training stride 16 and evaluation stride 32, except Hopper, which uses length 4 and strides 2 and 4, respectively. The transition protocol evaluates individual transitions and has no carried sequence state. Models predict the same training-standardized next-state displacement. Native-time open-loop replay is reported separately from teacher-forced displacement accuracy.

Primary development searches use learning rates $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$ and weight decays $\{0, 10^{-4}\}$, seeds 1–3, at most 50 epochs and patience 10. Final fits use at most 100 epochs and patience 20. The shared optimizer is unfused AdamW with cosine decay, gradient clipping 1, and batch size 256. The configured depth is two for sequence and four for transition models where the architecture exposes this setting; time GRU uses one recurrent cell. The WSP reference width is 128 and conditioner rank is 16. Learned comparator widths are parameter-matched within the declared 5% tolerance. Closed-form and parameter-free controls are not made artificially equal in size. The tables give achieved counts and selected widths rather than treating parameter matching as equal compute.

Primary execution uses independent jobs on eight NVIDIA B200 GPUs, PyTorch 2.9.1/CUDA 13.0, BF16 mixed precision, four CPU threads per task, and unfused AdamW. Matrix-multiplication TF32 is disabled; cuDNN TF32 is allowed. Sensitive recurrence computations retain the implementation's FP32 policy. Reference-versus-optimized runtime probes passed for both protocols. The recorded task-code fingerprint starts with `1057cc2e301c`.

Transformer means the causal sequence Transformer in sequence tables and the local Transformer baseline in transition tables. The transition *Token Transformer* is a separate local tokenization control. The two diagonal SSMs and TGNN4I-style probe are common-task adaptations, not native benchmark reproductions. WSP ablations inherit WSP's selected optimizer

setting and keep their recorded parameter tensors; disabling a component does not necessarily remove its stored parameters. In the accuracy tables, d is selected model width; a dash denotes a closed-form or parameter-free control. Validation columns describe the ten final fits, not the three-seed scores used for model selection.

Table A.1: Complete primary sequence results, including all component ablations and simple controls. Test and validation NRMSE are mean ± SD over ten final seeds; parameter counts are exact.

Model	Test NRMSE	Val. NRMSE	d	Parameters
Damped oscillator				
WSP carry	1.8352 ± 0.0056	0.1653 ± 0.0064	128	158,976
WSP reset	1.8389 ± 0.0174	0.0369 ± 0.0051	128	158,976
WSP decay only	1.8440 ± 0.0278	0.1715 ± 0.0074	128	158,976
WSP rotation only	2.0073 ± 0.1222	0.1107 ± 0.0041	128	158,976
WSP cosine only	1.9240 ± 0.0390	0.0818 ± 0.0078	128	158,976
WSP no conditioner	1.7834 ± 0.0067	0.1346 ± 0.0035	128	158,976
WSP no zero-time	1.8308 ± 0.0087	0.1631 ± 0.0098	128	158,976
Transformer	1.8312 ± 0.1966	0.3885 ± 0.1081	94	158,014
Time GRU	1.9479 ± 0.0344	0.2508 ± 0.0561	161	159,070
Diag. SSM (cond.)	1.8211 ± 0.0053	0.1502 ± 0.0115	134	159,192
Diag. SSM (global)	1.7986 ± 0.0057	0.1236 ± 0.0026	140	159,600
TGNN4I-style probe	1.8428 ± 0.0233	0.0708 ± 0.0100	53	159,636
Zero-time MLP	1.8225 ± 0.0733	0.1681 ± 0.0310	390	158,730
Zero-time ridge	1.6078 ± 0.0251	0.1116 ± 0.0021	–	80
Zero displacement	2.4763 ± 0.0000	0.8916 ± 0.0000	–	0
Constant velocity	1.1148 ± 0.0000	0.1880 ± 0.0000	–	0
Training mean	2.4763 ± 0.0000	0.8916 ± 0.0000	–	0
HalfCheetah				
WSP carry	0.7130 ± 0.0109	0.2382 ± 0.0130	128	163,456
WSP reset	1.0273 ± 0.0531	0.1475 ± 0.0092	128	163,456
WSP decay only	0.7034 ± 0.0056	0.1917 ± 0.0022	128	163,456
WSP rotation only	1.4350 ± 0.1399	0.2488 ± 0.0038	128	163,456
WSP cosine only	1.0276 ± 0.0522	0.2034 ± 0.0088	128	163,456
WSP no conditioner	0.6951 ± 0.0061	0.2282 ± 0.0032	128	163,456

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Table A.1 (continued)

Model	Test NRMSE	Val. NRMSE	d	Parameters
WSP no zero-time	0.6881 ± 0.0044	0.2338 ± 0.0080	128	163,456
Transformer	1.0985 ± 0.0693	0.5761 ± 0.0277	95	164,635
Time GRU	0.8602 ± 0.0133	0.4159 ± 0.0253	160	162,737
Diag. SSM (cond.)	0.7307 ± 0.0350	0.2522 ± 0.0032	134	163,882
Diag. SSM (global)	0.7579 ± 0.0077	0.3452 ± 0.0018	140	164,500
TGNN4I-style probe	1.4197 ± 0.0806	0.2183 ± 0.0071	53	161,491
Zero-time MLP	0.9990 ± 0.0576	0.3775 ± 0.0178	379	163,349
Zero-time ridge	0.9826 ± 0.0076	0.3914 ± 0.0002	–	4,080
Zero displacement	1.1827 ± 0.0000	0.9756 ± 0.0000	–	0
Constant velocity	3.0599 ± 0.0000	1.4655 ± 0.0000	–	0
Training mean	1.1824 ± 0.0000	0.9755 ± 0.0000	–	0
Hopper				
WSP carry	3.1552 ± 0.0292	0.1939 ± 0.0115	128	161,536
WSP reset	3.1608 ± 0.0281	0.0827 ± 0.0017	128	161,536
WSP decay only	2.7276 ± 0.0095	0.1614 ± 0.0074	128	161,536
WSP rotation only	3.3730 ± 0.0297	0.2007 ± 0.0142	128	161,536
WSP cosine only	3.1453 ± 0.0351	0.1221 ± 0.0061	128	161,536
WSP no conditioner	3.3296 ± 0.0202	0.2165 ± 0.0069	128	161,536
WSP no zero-time	3.1315 ± 0.0269	0.2066 ± 0.0223	128	161,536
Transformer	3.3998 ± 0.0510	0.5064 ± 0.0343	94	159,894
Time GRU	3.3724 ± 0.0309	0.3951 ± 0.0153	161	162,299
Diag. SSM (cond.)	3.1946 ± 0.0283	0.2599 ± 0.0101	134	161,872
Diag. SSM (global)	3.3624 ± 0.0174	0.3618 ± 0.0022	140	162,400
TGNN4I-style probe	3.3025 ± 0.0268	0.1526 ± 0.0119	53	160,696
Zero-time MLP	3.1178 ± 0.0221	0.3458 ± 0.0197	384	161,664
Zero-time ridge	3.1162 ± 0.0001	$0.4019 \pm 3.56 \times 10^{-6}$	–	1,650
Zero displacement	3.9583 ± 0.0000	0.8447 ± 0.0000	–	0
Constant velocity	3.6563 ± 0.0000	0.7297 ± 0.0000	–	0
Training mean	3.8964 ± 0.0000	0.8116 ± 0.0000	–	0
Reacher				
WSP carry	1.2442 ± 0.0075	0.2074 ± 0.0062	128	161,152
WSP reset	1.3011 ± 0.0134	0.1027 ± 0.0063	128	161,152
WSP decay only	1.2801 ± 0.0262	0.2465 ± 0.0046	128	161,152
WSP rotation only	1.5704 ± 0.1139	0.1594 ± 0.0040	128	161,152

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Table A.1 (continued)

Model	Test NRMSE	Val. NRMSE	d	Parameters
WSP cosine only	1.3375 ± 0.0233	0.1863 ± 0.0078	128	161,152
WSP no conditioner	1.2308 ± 0.0077	0.1858 ± 0.0051	128	161,152
WSP no zero-time	1.2235 ± 0.0051	0.2053 ± 0.0071	128	161,152
Transformer	1.3914 ± 0.0451	0.5158 ± 0.0415	94	159,612
Time GRU	1.3155 ± 0.0107	0.4967 ± 0.0304	161	161,815
Diag. SSM (cond.)	1.2460 ± 0.0129	0.2007 ± 0.0070	134	161,470
Diag. SSM (global)	1.3023 ± 0.0022	0.4694 ± 0.0043	140	161,980
TGNN4I-style probe	1.4577 ± 0.0932	0.2251 ± 0.0161	53	160,537
Zero-time MLP	1.3407 ± 0.0527	0.4535 ± 0.0412	385	161,315
Zero-time ridge	1.1574 ± 0.0101	0.6864 ± 0.0001	–	1,300
Zero displacement	1.6393 ± 0.0000	0.8287 ± 0.0000	–	0
Constant velocity	3.4622 ± 0.0000	1.2050 ± 0.0000	–	0
Training mean	1.6393 ± 0.0000	0.8287 ± 0.0000	–	0
Δt-*irrelevant*				
WSP carry	0.0169 ± 0.0006	0.0202 ± 0.0010	128	160,128
WSP reset	0.3653 ± 0.0453	0.0273 ± 0.0061	128	160,128
WSP decay only	0.0287 ± 0.0019	0.0239 ± 0.0016	128	160,128
WSP rotation only	0.3194 ± 0.0475	0.0265 ± 0.0011	128	160,128
WSP cosine only	0.0208 ± 0.0010	0.0204 ± 0.0004	128	160,128
WSP no conditioner	0.0480 ± 0.0015	0.0309 ± 0.0005	128	160,128
WSP no zero-time	0.0193 ± 0.0016	0.0369 ± 0.0039	128	160,128
Transformer	0.6455 ± 0.0462	0.3702 ± 0.0854	94	158,860
Time GRU	0.3273 ± 0.0367	0.2029 ± 0.0385	161	160,523
Diag. SSM (cond.)	0.0703 ± 0.0054	0.0424 ± 0.0033	134	160,398
Diag. SSM (global)	0.0721 ± 0.0022	0.0453 ± 0.0005	140	160,860
TGNN4I-style probe	0.7022 ± 0.0984	0.0504 ± 0.0077	53	160,113
Zero-time MLP	0.4758 ± 0.0652	0.1924 ± 0.0229	387	159,831
Zero-time ridge	0.3980 ± 0.0003	0.1562 ± 0.0002	–	540
Zero displacement	1.0051 ± 0.0000	1.0010 ± 0.0000	–	0
Constant velocity	6.3802 ± 0.0000	2.1268 ± 0.0000	–	0
Training mean	1.0051 ± 0.0000	1.0010 ± 0.0000	–	0

Table A.2: Complete primary transition-local results. All ten models are evaluated on each of the five environments; summaries are mean ± SD over ten final seeds.

Model	Test NRMSE	Val. NRMSE	d	Parameters
Damped oscillator				
WSP local	1.4045 ± 0.0791	0.0141 ± 0.0032	128	184,832
Transformer	1.4683 ± 0.0449	0.1460 ± 0.0137	73	183,522
Token Transformer	2.0296 ± 0.0798	0.0249 ± 0.0229	75	184,877
Diag. SSM (cond.)	1.5505 ± 0.0675	0.0093 ± 0.0007	101	185,840
Diag. SSM (global)	1.5370 ± 0.1223	0.0110 ± 0.0031	106	183,168
TGNN4I-style probe	1.7785 ± 0.1736	0.0168 ± 0.0015	40	182,400
Zero-time MLP	1.6828 ± 0.0547	0.1752 ± 0.0532	245	184,730
Zero-time ridge	1.5775 ± 0.0326	0.1119 ± 0.0029	–	80
Zero displacement	2.4469 ± 0.0000	0.8895 ± 0.0000	–	0
Training mean	2.4469 ± 0.0000	0.8895 ± 0.0000	–	0
HalfCheetah				
WSP local	1.2492 ± 0.0769	0.0866 ± 0.0057	128	189,312
Transformer	1.3253 ± 0.1827	0.3301 ± 0.0245	74	191,068
Token Transformer	1.2343 ± 0.2222	0.1261 ± 0.0307	75	189,767
Diag. SSM (cond.)	1.1621 ± 0.0862	0.0925 ± 0.0053	101	189,375
Diag. SSM (global)	3.3091 ± 0.7541	0.2045 ± 0.0122	107	190,353
TGNN4I-style probe	2.3516 ± 0.3020	0.1286 ± 0.0058	41	192,987
Zero-time MLP	1.3256 ± 0.2001	0.6234 ± 0.0880	242	188,760
Zero-time ridge	0.9808 ± 0.0066	0.3908 ± 0.0006	–	4,080
Zero displacement	1.1789 ± 0.0000	0.9733 ± 0.0000	–	0
Training mean	1.1787 ± 0.0000	0.9731 ± 0.0000	–	0
Hopper				
WSP local	4.2523 ± 0.0584	0.0696 ± 0.0054	128	187,392
Transformer	4.0147 ± 0.1084	0.2100 ± 0.0175	73	184,982
Token Transformer	3.8882 ± 0.2290	0.0634 ± 0.0058	75	187,736
Diag. SSM (cond.)	4.1180 ± 0.0952	0.0574 ± 0.0023	101	187,860
Diag. SSM (global)	4.8540 ± 0.1736	0.1931 ± 0.0075	107	188,748
TGNN4I-style probe	4.3728 ± 0.1536	0.0623 ± 0.0024	40	183,200
Zero-time MLP	4.1542 ± 0.1105	0.4092 ± 0.0389	244	188,124
Zero-time ridge	4.1555 ± 0.0001	$0.4200 \pm 5.99 \times 10^{-6}$	–	1,650

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Table A.2 (continued)

Model	Test NRMSE	Val. NRMSE	d	Parameters
Zero displacement	5.2895 ± 0.0000	0.8977 ± 0.0000	–	0
Training mean	5.1700 ± 0.0000	0.8506 ± 0.0000	–	0
Reacher				
WSP local	1.2656 ± 0.0368	0.0686 ± 0.0028	128	187,008
Transformer	1.3517 ± 0.0375	0.3087 ± 0.0149	73	184,763
Token Transformer	1.3044 ± 0.0951	0.0913 ± 0.0448	75	187,360
Diag. SSM (cond.)	1.1330 ± 0.0469	0.0651 ± 0.0032	101	187,557
Diag. SSM (global)	2.5322 ± 0.8831	0.2920 ± 0.0740	107	188,427
TGNN4I-style probe	1.5043 ± 0.1306	0.1105 ± 0.0569	40	183,080
Zero-time MLP	1.3400 ± 0.0548	0.5675 ± 0.0554	244	187,392
Zero-time ridge	1.1553 ± 0.0126	0.6876 ± 0.0007	–	1,300
Zero displacement	1.6403 ± 0.0000	0.8301 ± 0.0000	–	0
Training mean	1.6403 ± 0.0000	0.8301 ± 0.0000	–	0
Δt-*irrelevant*				
WSP local	0.4022 ± 0.1502	0.0039 ± 0.0001	128	185,984
Transformer	0.7275 ± 0.2488	0.1582 ± 0.0316	73	184,179
Token Transformer	0.3601 ± 0.0745	0.0054 ± 0.0005	75	186,156
Diag. SSM (cond.)	0.5772 ± 0.2359	0.0058 ± 0.0006	101	186,749
Diag. SSM (global)	1.1949 ± 0.1675	0.0066 ± 0.0002	107	187,571
TGNN4I-style probe	1.4296 ± 0.1899	0.0070 ± 0.0006	40	182,760
Zero-time MLP	0.1719 ± 0.0566	0.0671 ± 0.0207	244	185,440
Zero-time ridge	0.4020 ± 0.0100	0.1559 ± 0.0009	–	540
Zero displacement	1.0051 ± 0.0000	1.0010 ± 0.0000	–	0
Training mean	1.0051 ± 0.0000	1.0010 ± 0.0000	–	0

Table A.3: Complete primary PhysioNet results for all eleven models. These dense state-displacement predictions use the same prepared channels without observation masks; they are not mortality or mask-aware imputation scores.

Model	Test NRMSE	Val. NRMSE	d	Parameters
PhysioNet				
WSP carry	0.9086 ± 0.0017	0.9179 ± 0.0005	128	160,896
WSP reset	0.9257 ± 0.0008	0.9378 ± 0.0007	128	160,896
WSP decay only	0.9122 ± 0.0021	0.9210 ± 0.0020	128	160,896
Transformer	0.9092 ± 0.0015	0.9167 ± 0.0008	94	159,424
Time GRU	0.9012 ± 0.0010	0.9090 ± 0.0002	161	161,493
Diag. SSM (cond.)	0.9071 ± 0.0024	0.9175 ± 0.0006	134	161,202
Diag. SSM (global)	0.9078 ± 0.0003	0.9194 ± 0.0007	140	161,700
TGNN4I-style probe	0.9023 ± 0.0009	0.9129 ± 0.0004	53	160,431
Zero-time MLP	0.9250 ± 0.0012	0.9333 ± 0.0005	385	160,545
Zero displacement	0.9733 ± 0.0000	0.9858 ± 0.0000	–	0
Training mean	0.9733 ± 0.0000	0.9857 ± 0.0000	–	0

Appendix A.1. Paired final-seed comparisons

Differences pair the same ten final seeds. A positive $\Delta = \mathrm{NRMSE}_{\mathrm{baseline}} - \mathrm{NRMSE}_{\mathrm{WSP}}$ favors WSP. Intervals are percentile bootstrap 95% intervals from 20,000 paired resamples and are not simultaneous confidence intervals. Two-sided exact sign-flip tests are Holm-adjusted within the complete declared report family: 80 sequence comparisons, 45 transition comparisons, or 10 PhysioNet comparisons. WSP wins counts strictly lower seed-level error. The minimum attainable two-sided sign-flip p with ten pairs is $2/2^{10} = 0.001953125$; the broad family adjustments therefore limit resolved significance even when all ten seeds favor one model.

Table A.4: All 80 primary sequence comparisons against WSP carry. Holm correction is joint over this entire table.

Baseline	Δ	95% CI	Holm p	WSP wins
Damped oscillator				
WSP reset	0.0037	[−0.0053, 0.0129]	1.0000	6/10
WSP decay only	0.0088	[−0.0089, 0.0261]	1.0000	6/10
WSP rotation only	0.1721	[0.1079, 0.2520]	0.1562	10/10
WSP cosine only	0.0888	[0.0657, 0.1139]	0.1562	10/10
WSP no conditioner	−0.0518	[−0.0559, −0.0477]	0.1562	0/10
WSP no zero-time	−0.0044	[−0.0096, 0.0010]	1.0000	2/10
Transformer	−0.0039	[−0.1175, 0.1093]	1.0000	4/10
Time GRU	0.1127	[0.0928, 0.1343]	0.1562	10/10
Diag. SSM (cond.)	−0.0140	[−0.0191, −0.0089]	0.1562	1/10
Diag. SSM (global)	−0.0366	[−0.0404, −0.0329]	0.1562	0/10
TGNN4I-style probe	0.0076	[−0.0053, 0.0230]	1.0000	5/10
Zero-time MLP	−0.0127	[−0.0573, 0.0279]	1.0000	5/10
Zero-time ridge	−0.2274	[−0.2422, −0.2126]	0.1562	0/10
Zero displacement	0.6411	[0.6380, 0.6445]	0.1562	10/10
Constant velocity	−0.7204	[−0.7236, −0.7170]	0.1562	0/10
Training mean	0.6411	[0.6380, 0.6446]	0.1562	10/10
HalfCheetah				
WSP reset	0.3143	[0.2857, 0.3472]	0.1562	10/10
WSP decay only	−0.0097	[−0.0159, −0.0032]	0.2812	2/10
WSP rotation only	0.7220	[0.6420, 0.8062]	0.1562	10/10
WSP cosine only	0.3145	[0.2820, 0.3495]	0.1562	10/10
WSP no conditioner	−0.0179	[−0.0264, −0.0104]	0.1562	0/10
WSP no zero-time	−0.0250	[−0.0319, −0.0189]	0.1562	0/10
Transformer	0.3855	[0.3466, 0.4333]	0.1562	10/10
Time GRU	0.1472	[0.1361, 0.1569]	0.1562	10/10
Diag. SSM (cond.)	0.0177	[−0.0011, 0.0440]	1.0000	7/10
Diag. SSM (global)	0.0448	[0.0371, 0.0524]	0.1562	10/10
TGNN4I-style probe	0.7067	[0.6621, 0.7538]	0.1562	10/10
Zero-time MLP	0.2860	[0.2511, 0.3261]	0.1562	10/10
Zero-time ridge	0.2695	[0.2650, 0.2746]	0.1562	10/10
Zero displacement	0.4696	[0.4631, 0.4760]	0.1562	10/10

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Table A.4 (continued)

Baseline	Δ	95% CI	Holm p	WSP wins
Constant velocity	2.3469	[2.3404, 2.3531]	0.1562	10/10
Training mean	0.4694	[0.4629, 0.4756]	0.1562	10/10
Hopper				
WSP reset	0.0056	[−0.0185, 0.0318]	1.0000	5/10
WSP decay only	−0.4275	[−0.4483, −0.4099]	0.1562	0/10
WSP rotation only	0.2179	[0.2016, 0.2349]	0.1562	10/10
WSP cosine only	−0.0099	[−0.0383, 0.0136]	1.0000	5/10
WSP no conditioner	0.1744	[0.1511, 0.1952]	0.1562	10/10
WSP no zero-time	−0.0237	[−0.0424, −0.0036]	0.6016	3/10
Transformer	0.2446	[0.2126, 0.2719]	0.1562	10/10
Time GRU	0.2172	[0.1958, 0.2381]	0.1562	10/10
Diag. SSM (cond.)	0.0394	[0.0225, 0.0586]	0.1562	10/10
Diag. SSM (global)	0.2072	[0.1827, 0.2287]	0.1562	10/10
TGNN4I-style probe	0.1474	[0.1208, 0.1710]	0.1562	10/10
Zero-time MLP	−0.0374	[−0.0567, −0.0177]	0.1562	2/10
Zero-time ridge	−0.0390	[−0.0570, −0.0236]	0.1562	0/10
Zero displacement	0.8032	[0.7847, 0.8187]	0.1562	10/10
Constant velocity	0.5011	[0.4823, 0.5167]	0.1562	10/10
Training mean	0.7412	[0.7229, 0.7568]	0.1562	10/10
Reacher				
WSP reset	0.0569	[0.0476, 0.0659]	0.1562	10/10
WSP decay only	0.0360	[0.0224, 0.0486]	0.1562	9/10
WSP rotation only	0.3263	[0.2685, 0.3946]	0.1562	10/10
WSP cosine only	0.0933	[0.0770, 0.1094]	0.1562	10/10
WSP no conditioner	−0.0134	[−0.0211, −0.0060]	0.1562	1/10
WSP no zero-time	−0.0207	[−0.0262, −0.0152]	0.1562	0/10
Transformer	0.1472	[0.1170, 0.1748]	0.1562	10/10
Time GRU	0.0714	[0.0641, 0.0781]	0.1562	10/10
Diag. SSM (cond.)	0.0019	[−0.0084, 0.0107]	1.0000	6/10
Diag. SSM (global)	0.0581	[0.0529, 0.0627]	0.1562	10/10
TGNN4I-style probe	0.2135	[0.1611, 0.2703]	0.1562	10/10
Zero-time MLP	0.0965	[0.0655, 0.1315]	0.1562	10/10
Zero-time ridge	−0.0868	[−0.0971, −0.0795]	0.1562	0/10
Zero displacement	0.3951	[0.3904, 0.3992]	0.1562	10/10

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Table A.4 (continued)

Baseline	Δ	95% CI	Holm p	WSP wins
Constant velocity	2.2180	[2.2132, 2.2220]	0.1562	10/10
Training mean	0.3951	[0.3905, 0.3993]	0.1562	10/10
Δt-*irrelevant*				
WSP reset	0.3484	[0.3197, 0.3737]	0.1562	10/10
WSP decay only	0.0118	[0.0107, 0.0128]	0.1562	10/10
WSP rotation only	0.3025	[0.2763, 0.3319]	0.1562	10/10
WSP cosine only	0.0039	[0.0032, 0.0045]	0.1562	10/10
WSP no conditioner	0.0311	[0.0299, 0.0322]	0.1562	10/10
WSP no zero-time	0.0024	[0.0013, 0.0037]	0.1562	10/10
Transformer	0.6286	[0.6010, 0.6549]	0.1562	10/10
Time GRU	0.3104	[0.2890, 0.3314]	0.1562	10/10
Diag. SSM (cond.)	0.0534	[0.0503, 0.0563]	0.1562	10/10
Diag. SSM (global)	0.0553	[0.0540, 0.0565]	0.1562	10/10
TGNN4I-style probe	0.6853	[0.6328, 0.7472]	0.1562	10/10
Zero-time MLP	0.4589	[0.4216, 0.4987]	0.1562	10/10
Zero-time ridge	0.3811	[0.3807, 0.3815]	0.1562	10/10
Zero displacement	0.9882	[0.9878, 0.9886]	0.1562	10/10
Constant velocity	6.3633	[6.3629, 6.3637]	0.1562	10/10
Training mean	0.9882	[0.9878, 0.9886]	0.1562	10/10

Table A.5: All 45 primary transition-local comparisons against WSP local. Holm correction is joint over this entire table.

Baseline	Δ	95% CI	Holm p	WSP wins
Damped oscillator				
Transformer	0.0639	[0.0068, 0.1218]	0.5801	7/10
Token Transformer	0.6251	[0.5683, 0.6820]	0.0879	10/10
Diag. SSM (cond.)	0.1461	[0.0913, 0.2000]	0.0879	10/10
Diag. SSM (global)	0.1326	[0.0548, 0.2030]	0.1465	9/10
TGNN4I-style probe	0.3741	[0.2811, 0.4917]	0.0879	10/10
Zero-time MLP	0.2784	[0.2155, 0.3338]	0.0879	10/10
Zero-time ridge	0.1731	[0.1217, 0.2231]	0.0879	10/10

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Table A.5 (continued)

Baseline	Δ	95% CI	Holm p	WSP wins
Zero displacement	1.0424	[0.9952, 1.0867]	0.0879	10/10
Training mean	1.0424	[0.9950, 1.0875]	0.0879	10/10
HalfCheetah				
Transformer	0.0760	[−0.0459, 0.1972]	1.0000	7/10
Token Transformer	−0.0149	[−0.1401, 0.1360]	1.0000	4/10
Diag. SSM (cond.)	−0.0871	[−0.1471, −0.0183]	0.4297	2/10
Diag. SSM (global)	2.0598	[1.6445, 2.4910]	0.0879	10/10
TGNN4I-style probe	1.1023	[0.9497, 1.2598]	0.0879	10/10
Zero-time MLP	0.0764	[−0.0224, 0.1845]	1.0000	7/10
Zero-time ridge	−0.2684	[−0.3081, −0.2204]	0.0879	0/10
Zero displacement	−0.0703	[−0.1097, −0.0201]	0.3047	1/10
Training mean	−0.0706	[−0.1100, −0.0206]	0.3047	1/10
Hopper				
Transformer	−0.2376	[−0.3198, −0.1518]	0.0879	1/10
Token Transformer	−0.3641	[−0.4756, −0.2612]	0.0879	0/10
Diag. SSM (cond.)	−0.1344	[−0.2069, −0.0602]	0.2188	2/10
Diag. SSM (global)	0.6017	[0.5124, 0.6864]	0.0879	10/10
TGNN4I-style probe	0.1204	[0.0182, 0.2192]	0.5469	6/10
Zero-time MLP	−0.0982	[−0.1713, −0.0018]	0.5801	2/10
Zero-time ridge	−0.0968	[−0.1304, −0.0616]	0.0879	0/10
Zero displacement	1.0371	[1.0039, 1.0733]	0.0879	10/10
Training mean	0.9177	[0.8844, 0.9523]	0.0879	10/10
Reacher				
Transformer	0.0862	[0.0605, 0.1106]	0.0879	10/10
Token Transformer	0.0388	[−0.0143, 0.0985]	1.0000	6/10
Diag. SSM (cond.)	−0.1325	[−0.1645, −0.1011]	0.0879	0/10
Diag. SSM (global)	1.2667	[0.8340, 1.8347]	0.0879	10/10
TGNN4I-style probe	0.2387	[0.1450, 0.3317]	0.0879	9/10
Zero-time MLP	0.0744	[0.0367, 0.1183]	0.0879	9/10
Zero-time ridge	−0.1103	[−0.1329, −0.0907]	0.0879	0/10
Zero displacement	0.3747	[0.3510, 0.3944]	0.0879	10/10
Training mean	0.3747	[0.3511, 0.3942]	0.0879	10/10

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Table A.5 (continued)

Baseline	Δ	95% CI	Holm p	WSP wins
Δ*t-irrelevant*				
Transformer	0.3254	[0.1830, 0.4512]	0.0879	9/10
Token Transformer	−0.0421	[−0.1244, 0.0433]	1.0000	3/10
Diag. SSM (cond.)	0.1750	[−0.0178, 0.3681]	0.8750	7/10
Diag. SSM (global)	0.7927	[0.7064, 0.8940]	0.0879	10/10
TGNN4I-style probe	1.0275	[0.8657, 1.1933]	0.0879	10/10
Zero-time MLP	−0.2302	[−0.3351, −0.1288]	0.0879	1/10
Zero-time ridge	−0.0002	[−0.0905, 0.0876]	1.0000	4/10
Zero displacement	0.6029	[0.5130, 0.6905]	0.0879	10/10
Training mean	0.6029	[0.5136, 0.6899]	0.0879	10/10

Table A.6: All ten primary PhysioNet comparisons against WSP carry. Holm correction is joint over this entire table.

Baseline	Δ	95% CI	Holm p	WSP wins
PhysioNet				
WSP reset	0.0172	[0.0159, 0.0184]	0.0195	10/10
WSP decay only	0.0036	[0.0020, 0.0051]	0.0234	9/10
Transformer	0.0006	[−0.0008, 0.0017]	0.4121	7/10
Time GRU	−0.0073	[−0.0084, −0.0062]	0.0195	0/10
Diag. SSM (cond.)	−0.0015	$[-0.0029, -4.13\times10^{-5}]$	0.2812	3/10
Diag. SSM (global)	−0.0008	[−0.0018, 0.0003]	0.3711	3/10
TGNN4I-style probe	−0.0063	[−0.0073, −0.0053]	0.0195	0/10
Zero-time MLP	0.0164	[0.0151, 0.0179]	0.0195	10/10
Zero displacement	0.0648	[0.0638, 0.0658]	0.0195	10/10
Training mean	0.0647	[0.0638, 0.0657]	0.0195	10/10

Appendix A.2. Elapsed-time interventions

For timing interventions, states, actions and targets are held fixed. Shuffling permutes gaps within an evaluated batch; constant time replaces them by the batch mean. Positive intervention-minus-correct NRMSE denotes deterioration, not an accuracy percentage. These paired intervals describe training-seed variability on the same targets. They do not distinguish a

learned timing dependence from every possible distribution shift induced by changing time inputs. In particular, a zero-time test against unchanged nonzero-time targets also probes the imposed zero-displacement boundary.

Table A.7: WSP carry timing effects on all six primary sequence datasets. Mean and SD refer to paired intervention-minus-correct NRMSE over ten seeds.

Environment	Time input	Effect ± SD	95% CI
Damped oscillator	Constant	0.0067 ± 0.0040	[0.0044, 0.0091]
Damped oscillator	Shuffled	0.1026 ± 0.0025	[0.1013, 0.1042]
HalfCheetah	Constant	0.0397 ± 0.0088	[0.0342, 0.0445]
HalfCheetah	Shuffled	0.1005 ± 0.0066	[0.0964, 0.1041]
Hopper	Constant	0.0214 ± 0.0211	[0.0091, 0.0338]
Hopper	Shuffled	0.1828 ± 0.0380	[0.1604, 0.2050]
Reacher	Constant	0.0513 ± 0.0025	[0.0498, 0.0527]
Reacher	Shuffled	0.0809 ± 0.0029	[0.0790, 0.0824]
Δt-irrelevant	Constant	-0.0029 ± 0.0003	[−0.0031, −0.0028]
Δt-irrelevant	Shuffled	-0.0001 ± 0.0003	[−0.0002, 0.0001]
PhysioNet	Constant	0.0040 ± 0.0023	[0.0026, 0.0053]
PhysioNet	Shuffled	0.0059 ± 0.0024	[0.0046, 0.0073]

Appendix A.3. Native-horizon open-loop replay

Replay feeds predicted states back while retaining the recorded actions and elapsed intervals. It is an open-loop prediction diagnostic, not closed-loop policy return. At horizon H, error aggregates all first H predicted transitions and is normalized by training-state standard deviations; it is not endpoint-only error or the displacement NRMSE in the accuracy tables. Hopper has only four-transition windows; other datasets support horizons 4, 8, 16 and 32. PhysioNet steps are successive irregular records, not equally spaced physical durations.

The selected models below include the plotted comparators, reset-state and simple state controls. Constant velocity is omitted from replay: initialization from a single state provides no prior difference to estimate velocity, so that replay implementation coincides with zero displacement. Its teacher-forced one-step result remains in Table A.1.

Table A.8: Native-horizon replay NRMSE (mean ± sample SD over ten seeds). Dashes denote horizons beyond Hopper's native window; only models in each dataset's manifest are shown.

Model	$H = 4$	$H = 8$	$H = 16$	$H = 32$
Damped oscillator				
WSP carry	0.377 ± 0.002	0.601 ± 0.003	0.870 ± 0.003	0.842 ± 0.002
WSP reset	0.362 ± 0.006	0.570 ± 0.007	0.837 ± 0.013	0.860 ± 0.021
WSP decay only	0.332 ± 0.022	0.535 ± 0.030	0.786 ± 0.027	0.779 ± 0.014
Transformer	0.368 ± 0.048	0.600 ± 0.069	0.893 ± 0.065	0.940 ± 0.035
Time GRU	0.402 ± 0.010	0.637 ± 0.014	0.917 ± 0.018	0.906 ± 0.043
Diag. SSM (cond.)	0.371 ± 0.001	0.590 ± 0.002	0.847 ± 0.005	0.814 ± 0.007
Diag. SSM (global)	0.366 ± 0.002	0.582 ± 0.003	0.845 ± 0.004	0.827 ± 0.004
TGNN4I-style probe	0.365 ± 0.005	0.578 ± 0.010	0.847 ± 0.020	0.860 ± 0.026
Zero-time ridge	0.332 ± 0.005	0.544 ± 0.007	0.831 ± 0.009	0.880 ± 0.007
Zero displacement	0.531 ± 0.000	0.822 ± 0.000	1.020 ± 0.000	0.913 ± 0.000
HalfCheetah				
WSP carry	0.625 ± 0.013	0.649 ± 0.022	0.673 ± 0.036	0.704 ± 0.064
WSP reset	0.908 ± 0.051	0.988 ± 0.080	1.085 ± 0.142	1.193 ± 0.211
WSP decay only	0.635 ± 0.038	0.687 ± 0.082	0.743 ± 0.169	0.821 ± 0.323
Transformer	0.906 ± 0.098	0.942 ± 0.103	0.972 ± 0.105	0.999 ± 0.111
Time GRU	0.679 ± 0.018	0.682 ± 0.019	0.686 ± 0.020	0.695 ± 0.021
Diag. SSM (cond.)	0.516 ± 0.004	0.522 ± 0.004	0.537 ± 0.007	0.647 ± 0.154
Diag. SSM (global)	0.587 ± 0.014	0.613 ± 0.013	0.646 ± 0.012	0.685 ± 0.024
TGNN4I-style probe	1.289 ± 0.122	1.404 ± 0.178	1.510 ± 0.240	1.589 ± 0.300
Zero-time ridge	0.984 ± 0.026	1.080 ± 0.037	1.134 ± 0.046	1.169 ± 0.050
Zero displacement	0.830 ± 0.000	0.893 ± 0.000	0.941 ± 0.000	0.972 ± 0.000
Hopper				
WSP carry	1.366 ± 0.008	–	–	–
WSP reset	1.297 ± 0.021	–	–	–
WSP decay only	1.065 ± 0.007	–	–	–
Transformer	1.353 ± 0.027	–	–	–
Time GRU	1.388 ± 0.023	–	–	–
Diag. SSM (cond.)	1.297 ± 0.012	–	–	–
Diag. SSM (global)	1.363 ± 0.013	–	–	–

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Table A.8 (continued)

Model	$H = 4$	$H = 8$	$H = 16$	$H = 32$
TGNN4I-style probe	1.331 ± 0.037	–	–	–
Zero-time ridge	$1.316 \pm 4.54 \times 10^{-5}$	–	–	–
Zero displacement	1.631 ± 0.000	–	–	–
Reacher				
WSP carry	0.850 ± 0.004	0.907 ± 0.006	0.951 ± 0.007	0.971 ± 0.010
WSP reset	0.880 ± 0.014	0.943 ± 0.016	0.977 ± 0.018	0.994 ± 0.022
WSP decay only	0.879 ± 0.012	0.939 ± 0.017	0.990 ± 0.028	1.082 ± 0.107
Transformer	0.915 ± 0.045	0.982 ± 0.061	1.029 ± 0.059	1.066 ± 0.063
Time GRU	0.759 ± 0.010	0.809 ± 0.012	0.858 ± 0.015	0.885 ± 0.019
Diag. SSM (cond.)	0.786 ± 0.014	0.849 ± 0.015	0.935 ± 0.028	1.018 ± 0.050
Diag. SSM (global)	0.810 ± 0.013	0.878 ± 0.017	0.977 ± 0.016	1.130 ± 0.095
TGNN4I-style probe	0.943 ± 0.082	0.988 ± 0.090	1.020 ± 0.097	1.036 ± 0.105
Zero-time ridge	0.699 ± 0.002	0.709 ± 0.003	0.719 ± 0.004	0.721 ± 0.004
Zero displacement	0.964 ± 0.000	1.024 ± 0.000	1.091 ± 0.000	1.148 ± 0.000
Δt*-irrelevant*				
WSP carry	0.008 ± 0.001	0.008 ± 0.001	0.007 ± 0.001	$0.006 \pm 4.79 \times 10^{-4}$
WSP reset	0.098 ± 0.012	0.128 ± 0.019	0.164 ± 0.031	0.201 ± 0.044
WSP decay only	0.017 ± 0.001	0.018 ± 0.002	0.017 ± 0.001	0.014 ± 0.001
Transformer	0.161 ± 0.050	0.219 ± 0.068	0.278 ± 0.073	0.315 ± 0.064
Time GRU	0.106 ± 0.009	0.130 ± 0.014	0.155 ± 0.022	0.178 ± 0.032
Diag. SSM (cond.)	0.021 ± 0.002	0.024 ± 0.002	0.028 ± 0.002	0.035 ± 0.003
Diag. SSM (global)	$0.020 \pm 4.44 \times 10^{-4}$	$0.023 \pm 4.55 \times 10^{-4}$	$0.024 \pm 3.97 \times 10^{-4}$	$0.025 \pm 2.97 \times 10^{-4}$
TGNN4I-style probe	0.145 ± 0.027	0.196 ± 0.044	0.255 ± 0.066	0.307 ± 0.088
Zero-time ridge	$0.106 \pm 1.57 \times 10^{-4}$	$0.123 \pm 2.00 \times 10^{-4}$	$0.140 \pm 2.90 \times 10^{-4}$	$0.155 \pm 4.06 \times 10^{-4}$
Zero displacement	0.451 ± 0.000	0.699 ± 0.000	1.016 ± 0.000	1.299 ± 0.000
PhysioNet				
WSP carry	0.944 ± 0.001	0.968 ± 0.003	0.958 ± 0.005	0.933 ± 0.007
WSP reset	0.949 ± 0.001	0.973 ± 0.001	0.965 ± 0.002	0.942 ± 0.006
WSP decay only	0.948 ± 0.003	0.972 ± 0.003	0.963 ± 0.005	0.944 ± 0.011
Transformer	0.950 ± 0.002	0.968 ± 0.002	0.953 ± 0.003	0.930 ± 0.009
Time GRU	0.944 ± 0.001	0.960 ± 0.001	0.944 ± 0.001	0.915 ± 0.002
Diag. SSM (cond.)	0.945 ± 0.001	0.969 ± 0.003	0.955 ± 0.004	0.925 ± 0.007
Diag. SSM (global)	0.948 ± 0.003	0.971 ± 0.002	0.956 ± 0.002	0.921 ± 0.002
TGNN4I-style probe	0.946 ± 0.002	0.963 ± 0.002	0.947 ± 0.001	0.921 ± 0.003

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Table A.8 (continued)

Model	$H = 4$	$H = 8$	$H = 16$	$H = 32$
Zero displacement	0.984 ± 0.000	1.006 ± 0.000	0.992 ± 0.000	0.958 ± 0.000

Appendix A.4. Accuracy-matched PhysioNet inference workloads

Inference timings are implementation-specific measurements on an idle, exclusive NVIDIA B200, using BF16, 30 warm-up iterations and 200 timed repeats with interleaved model order. They exclude training and should not be interpreted as optimized-kernel bounds. Native evaluation length is 32; timings at other lengths are not substituted for the model's evaluated accuracy. Throughput counts transitions, so a batch of B length-32 sequences contains $32B$ transitions. Peak allocated memory is in MiB, includes the loaded model and workload, and is not an isolated state-size measurement.

The matching rule is $100|E_{\text{candidate}} - E_{\text{WSP}}|/E_{\text{WSP}} \leq 1$ using mean validation NRMSE. This point-estimate rule is not an equivalence test. The five PhysioNet comparators below meet it at the native length; neither primary controlled-task protocol has a matched pair. Matching uses validation rather than test scores. WSP has mean validation NRMSE 0.917887. The candidate values are listed in Table A.3; timing comparisons do not alter checkpoint selection.

Table A.9: All twenty matched PhysioNet pairs, with WSP reference rows. Throughput is mean ± SD across ten final-seed checkpoints. Ratio is WSP/candidate throughput (below one favors the candidate); memory saving is $100(1 - M_{\text{WSP}}/M_{\text{candidate}})$, so negative values mean WSP uses more memory.

Model	10^6 transitions/s	Peak MiB	Ratio	Saving (%)
Batch size 1; sequence length 32				
WSP carry	0.0010 ± 0.0001	10.077	1.000	–
Transformer	0.0238 ± 0.0002	10.101	0.043	0.24
Time GRU	0.0188 ± 0.0008	10.085	0.055	0.08
Diag. SSM (cond.)	$0.0015 \pm 5.57 \times 10^{-6}$	10.095	0.688	0.17
Diag. SSM (global)	0.0016 ± 0.0001	10.096	0.627	0.18

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Table A.9 (continued)

Model	10^6 transitions/s	Peak MiB	Ratio	Saving (%)
TGNN4I-style probe	0.0015 ± 0.0001	10.080	0.672	0.02
Batch size 16; sequence length 32				
WSP carry	0.0154 ± 0.0010	10.338	1.000	–
Transformer	0.3469 ± 0.0067	11.170	0.044	7.45
Time GRU	0.1282 ± 0.0012	10.312	0.120	-0.25
Diag. SSM (cond.)	0.0210 ± 0.0015	10.305	0.730	-0.32
Diag. SSM (global)	0.0238 ± 0.0018	10.314	0.644	-0.23
TGNN4I-style probe	0.0228 ± 0.0016	10.179	0.674	-1.56
Batch size 256; sequence length 32				
WSP carry	0.2230 ± 0.0088	14.566	1.000	–
Transformer	4.9884 ± 0.1716	29.691	0.045	50.94
Time GRU	4.2797 ± 0.1595	14.087	0.052	-3.40
Diag. SSM (cond.)	0.3276 ± 0.0207	13.780	0.681	-5.70
Diag. SSM (global)	0.3711 ± 0.0211	13.919	0.601	-4.65
TGNN4I-style probe	0.3325 ± 0.0113	11.839	0.671	-23.03
Batch size 4,096; sequence length 32				
WSP carry	3.7245 ± 0.2695	85.122	1.000	–
Transformer	21.8439 ± 0.0855	327.441	0.171	74.00
Time GRU	74.1494 ± 4.2781	75.744	0.050	-12.38
Diag. SSM (cond.)	5.7632 ± 0.3968	70.323	0.646	-21.04
Diag. SSM (global)	6.1850 ± 0.4236	72.605	0.602	-17.24
TGNN4I-style probe	5.8120 ± 0.4006	38.881	0.641	-118.93

Appendix A.5. Locked primary optimizer settings

Each cell below is selected before final testing by minimum mean development validation NRMSE. Exact ties prefer lower learning rate, then lower weight decay, then trial identifier. There are 70 main-suite locks and seven independent PhysioNet locks. The complete manifests and semantic lock hashes verify against the archived records. WSP ablations reuse the WSP carry setting for their dataset; fixed and closed-form controls do not receive a neural-optimizer search. For ridge regression, the training-fit regularization grid is $\{10^{-6}, 10^{-4}, 10^{-2}, 1, 100\}$ with validation selection.

Table A.10: All 77 primary selected optimizer settings. The validation mean ± SD uses development seeds 1–3 only. LR and WD denote learning rate and weight decay.

Model	LR	WD	Dev. val. NRMSE
Damped oscillator — sequence			
WSP carry	10^{-3}	0	0.1701 ± 0.0144
Transformer	10^{-3}	10^{-4}	0.4216 ± 0.0574
Time GRU	10^{-3}	0	0.3096 ± 0.0710
Diag. SSM (cond.)	10^{-3}	10^{-4}	0.1517 ± 0.0059
Diag. SSM (global)	10^{-3}	10^{-4}	0.1269 ± 0.0046
TGNN4I-style probe	10^{-3}	10^{-4}	0.0706 ± 0.0074
Zero-time MLP	10^{-4}	0	0.1956 ± 0.0165
HalfCheetah — sequence			
WSP carry	10^{-3}	0	0.2448 ± 0.0129
Transformer	10^{-3}	10^{-4}	0.5128 ± 0.0569
Time GRU	3×10^{-4}	10^{-4}	0.4092 ± 0.0092
Diag. SSM (cond.)	10^{-3}	10^{-4}	0.2681 ± 0.0022
Diag. SSM (global)	10^{-3}	0	0.3486 ± 0.0008
TGNN4I-style probe	10^{-3}	10^{-4}	0.2297 ± 0.0091
Zero-time MLP	10^{-4}	0	0.3917 ± 0.0251
Hopper — sequence			
WSP carry	10^{-3}	0	0.2179 ± 0.0167
Transformer	10^{-3}	0	0.4890 ± 0.0403
Time GRU	3×10^{-4}	10^{-4}	0.3954 ± 0.0295
Diag. SSM (cond.)	10^{-3}	10^{-4}	0.2579 ± 0.0002
Diag. SSM (global)	10^{-3}	0	0.3624 ± 0.0010
TGNN4I-style probe	10^{-3}	0	0.1632 ± 0.0035
Zero-time MLP	10^{-3}	0	0.3297 ± 0.0107
Reacher — sequence			
WSP carry	10^{-3}	10^{-4}	0.2109 ± 0.0102
Transformer	10^{-3}	0	0.5633 ± 0.0292
Time GRU	10^{-3}	10^{-4}	0.5267 ± 0.0295
Diag. SSM (cond.)	10^{-3}	0	0.2042 ± 0.0047
Diag. SSM (global)	10^{-3}	10^{-4}	0.4815 ± 0.0017

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Table A.10 (continued)

Model	LR	WD	Dev. val. NRMSE
TGNN4I-style probe	10^{-3}	10^{-4}	0.2207 ± 0.0028
Zero-time MLP	10^{-3}	0	0.4564 ± 0.0328
Δt-irrelevant — sequence			
WSP carry	10^{-3}	0	0.0252 ± 0.0003
Transformer	10^{-3}	0	0.3735 ± 0.1677
Time GRU	3×10^{-4}	10^{-4}	0.2172 ± 0.0350
Diag. SSM (cond.)	10^{-3}	0	0.0530 ± 0.0029
Diag. SSM (global)	10^{-3}	0	0.0583 ± 0.0001
TGNN4I-style probe	10^{-3}	0	0.0610 ± 0.0108
Zero-time MLP	10^{-4}	0	0.1856 ± 0.0191
Damped oscillator — transition			
WSP local	10^{-4}	0	0.0151 ± 0.0013
Transformer	10^{-3}	0	0.1359 ± 0.0102
Token Transformer	10^{-4}	0	0.0148 ± 0.0019
Diag. SSM (cond.)	10^{-3}	0	0.0097 ± 0.0012
Diag. SSM (global)	3×10^{-4}	10^{-4}	0.0104 ± 0.0019
TGNN4I-style probe	3×10^{-4}	10^{-4}	0.0136 ± 0.0007
Zero-time MLP	10^{-4}	0	0.1767 ± 0.1062
HalfCheetah — transition			
WSP local	10^{-3}	0	0.0896 ± 0.0009
Transformer	10^{-4}	0	0.3356 ± 0.0308
Token Transformer	3×10^{-4}	0	0.1206 ± 0.0228
Diag. SSM (cond.)	10^{-3}	0	0.0955 ± 0.0063
Diag. SSM (global)	10^{-3}	0	0.2130 ± 0.0024
TGNN4I-style probe	10^{-3}	0	0.1268 ± 0.0009
Zero-time MLP	10^{-4}	0	0.6039 ± 0.0460
Hopper — transition			
WSP local	10^{-3}	10^{-4}	0.0696 ± 0.0007
Transformer	10^{-3}	10^{-4}	0.2149 ± 0.0132
Token Transformer	10^{-3}	10^{-4}	0.0715 ± 0.0125
Diag. SSM (cond.)	10^{-3}	0	0.0676 ± 0.0035
Diag. SSM (global)	10^{-3}	10^{-4}	0.2059 ± 0.0070
TGNN4I-style probe	10^{-3}	0	0.0744 ± 0.0016

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Table A.10 (continued)

Model	LR	WD	Dev. val. NRMSE
Zero-time MLP	10^{-3}	0	0.3947 ± 0.0428
Reacher — transition			
WSP local	10^{-4}	0	0.0666 ± 0.0012
Transformer	10^{-4}	0	0.2988 ± 0.0089
Token Transformer	10^{-3}	10^{-4}	0.0907 ± 0.0195
Diag. SSM (cond.)	3×10^{-4}	10^{-4}	0.0643 ± 0.0012
Diag. SSM (global)	10^{-3}	10^{-4}	0.2878 ± 0.0730
TGNN4I-style probe	10^{-3}	0	0.0789 ± 0.0058
Zero-time MLP	3×10^{-4}	10^{-4}	0.5580 ± 0.0637
Δt*-irrelevant — transition*			
WSP local	10^{-3}	10^{-4}	0.0044 ± 0.0003
Transformer	10^{-3}	10^{-4}	0.2118 ± 0.0463
Token Transformer	3×10^{-4}	10^{-4}	0.0066 ± 0.0012
Diag. SSM (cond.)	10^{-3}	0	0.0056 ± 0.0001
Diag. SSM (global)	10^{-3}	0	0.0076 ± 0.0002
TGNN4I-style probe	10^{-3}	10^{-4}	0.0077 ± 0.0005
Zero-time MLP	10^{-4}	0	0.0818 ± 0.0233
PhysioNet — sequence			
WSP carry	10^{-3}	10^{-4}	0.9197 ± 0.0006
Transformer	10^{-3}	0	0.9163 ± 0.0007
Time GRU	10^{-3}	10^{-4}	0.9087 ± 0.0004
Diag. SSM (cond.)	10^{-3}	10^{-4}	0.9176 ± 0.0005
Diag. SSM (global)	10^{-3}	10^{-4}	0.9231 ± 0.0001
TGNN4I-style probe	10^{-3}	10^{-4}	0.9142 ± 0.0001
Zero-time MLP	10^{-3}	0	0.9338 ± 0.0004

Appendix B. Supplemental recurrent-competitor experiment

This separate experiment compares WSP carry, f-CRU, LRU, Phased LSTM, UnICORNN and the sequence Transformer on four datasets. All are common-task adaptations, not reproductions of their native benchmarks. Unlike the primary study, development uses seeds 1 and 2, three learning rates, and fixed weight decay 10^{-4}. Final seeds are again 101–110, so they are not extra independent replications of the primary results. The sequence

length, strides, epoch/patience limits and nominal parameter budget otherwise follow the recorded supplemental protocol. The original implementation runs independent jobs on eight B200 GPUs with unfused AdamW and a BF16 mixed-precision policy; f-CRU uses its FP32 path. Matrix-multiplication and cuDNN TF32 are disabled.

The supplemental task-code fingerprint starts with `166354f33edb`, not the primary fingerprint. PhysioNet also has a different recorded dataset fingerprint. Such a fingerprint includes metadata and shard names/sizes, so this alone does not establish different numerical samples. No cross-study seed pooling or cross-fingerprint PhysioNet inference is performed. Several overlapping seed-level scores are numerically identical; they are not counted twice as independent evidence. The tables below retain this experiment's own tuning lock, reference fit and multiplicity family.

Table B.11: Separate supplemental sequence experiment: all 24 model–dataset summaries (ten final seeds each). Mean ± SD and exact parameter counts are reported under this experiment's own tuning protocol.

Model	Test NRMSE	Val. NRMSE	d	Parameters
Damped oscillator				
WSP carry	1.8373 ± 0.0174	0.1771 ± 0.0113	128	158,976
Transformer	1.8602 ± 0.1459	0.3634 ± 0.1477	94	158,014
f-CRU	2.1431 ± 0.0643	0.0382 ± 0.0083	142	157,922
LRU	2.0255 ± 0.0421	0.4072 ± 0.1435	114	159,602
Phased LSTM	2.9202 ± 0.3486	0.4397 ± 0.0372	112	157,922
UnICORNN	2.5301 ± 0.0408	0.7235 ± 0.0226	388	158,694
HalfCheetah				
WSP carry	0.7108 ± 0.0110	0.2353 ± 0.0093	128	163,456
Transformer	1.0985 ± 0.0693	0.5761 ± 0.0277	95	164,635
f-CRU	1.2684 ± 0.0471	0.2279 ± 0.0098	142	162,907
LRU	0.9762 ± 0.0428	0.5365 ± 0.0443	114	163,607
Phased LSTM	2.6634 ± 0.6193	0.8347 ± 0.0213	110	162,927
UnICORNN	1.1821 ± 0.0003	0.9723 ± 0.0002	377	163,258
Δt*-irrelevant*				
WSP carry	0.0169 ± 0.0006	0.0202 ± 0.0010	128	160,128

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Table B.11 (continued)

Model	Test NRMSE	Val. NRMSE	d	Parameters
Transformer	0.6455 ± 0.0628	0.3681 ± 0.0834	94	158,860
f-CRU	0.4199 ± 0.1285	0.0407 ± 0.0009	142	159,204
LRU	0.4251 ± 0.0578	0.3139 ± 0.0802	114	160,632
Phased LSTM	4.1254 ± 1.0509	0.3494 ± 0.0264	112	160,614
UnICORNN	0.9370 ± 0.0010	0.9353 ± 0.0008	385	159,781
PhysioNet				
WSP carry	0.9086 ± 0.0017	0.9179 ± 0.0005	128	160,896
Transformer	0.9091 ± 0.0016	0.9166 ± 0.0008	94	159,424
f-CRU	0.9091 ± 0.0019	0.9149 ± 0.0009	142	160,060
LRU	0.9055 ± 0.0018	0.9090 ± 0.0003	114	161,320
Phased LSTM	0.9335 ± 0.0024	0.9476 ± 0.0025	112	161,962
UnICORNN	$0.9726 \pm 1.35\times10^{-5}$	$0.9851 \pm 1.29\times10^{-5}$	384	161,290

Table B.12: All twenty supplemental comparisons against that experiment's WSP carry. Positive differences favor WSP. Intervals and tests use the same paired-seed definitions as the primary study, with Holm correction over these twenty comparisons only.

Baseline	Δ	95% CI	Holm p	WSP wins
Damped oscillator				
Transformer	0.0229	[−0.0633, 0.1076]	1.0000	5/10
f-CRU	0.3058	[0.2623, 0.3463]	0.0391	10/10
LRU	0.1882	[0.1626, 0.2146]	0.0391	10/10
Phased LSTM	1.0828	[0.8927, 1.3051]	0.0391	10/10
UnICORNN	0.6927	[0.6695, 0.7202]	0.0391	10/10
HalfCheetah				
Transformer	0.3877	[0.3487, 0.4355]	0.0391	10/10
f-CRU	0.5576	[0.5296, 0.5828]	0.0391	10/10
LRU	0.2654	[0.2418, 0.2898]	0.0391	10/10
Phased LSTM	1.9526	[1.6098, 2.3365]	0.0391	10/10
UnICORNN	0.4713	[0.4646, 0.4774]	0.0391	10/10

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Table B.12 (continued)

Baseline	Δ	95% CI	Holm p	WSP wins
Δt-*irrelevant*				
Transformer	0.6286	[0.5927, 0.6656]	0.0391	10/10
f-CRU	0.4031	[0.3327, 0.4840]	0.0391	10/10
LRU	0.4083	[0.3726, 0.4405]	0.0391	10/10
Phased LSTM	4.1086	[3.5467, 4.7768]	0.0391	10/10
UnICORNN	0.9201	[0.9194, 0.9209]	0.0391	10/10
PhysioNet				
Transformer	0.0005	[−0.0007, 0.0017]	1.0000	7/10
f-CRU	0.0005	[−0.0009, 0.0021]	1.0000	6/10
LRU	−0.0031	[−0.0043, −0.0020]	0.0391	0/10
Phased LSTM	0.0249	[0.0236, 0.0262]	0.0391	10/10
UnICORNN	0.0641	[0.0631, 0.0651]	0.0391	10/10

Table B.13: Supplemental selected learning rates. Weight decay is 10^{-4} throughout; selection uses the mean validation NRMSE of development seeds 1 and 2.

Model	Oscillator	HalfCheetah	PhysioNet	Δt-irrelevant
WSP carry	3×10^{-4}	10^{-3}	10^{-3}	10^{-3}
f-CRU	10^{-3}	10^{-3}	10^{-3}	10^{-3}
LRU	10^{-3}	3×10^{-4}	10^{-3}	10^{-3}
Phased LSTM	10^{-3}	10^{-3}	10^{-3}	10^{-3}
UnICORNN	3×10^{-4}	10^{-3}	10^{-3}	10^{-3}
Transformer	3×10^{-4}	10^{-3}	10^{-3}	10^{-3}

References

[1] Z. Che, S. Purushotham, K. Cho, D. Sontag, Y. Liu, Recurrent neural networks for multivariate time series with missing values, Scientific Reports 8 (2018) 6085. doi:10.1038/s41598-018-24271-9.

[2] Y. Rubanova, R. T. Q. Chen, D. Duvenaud, Latent ordinary differential equations for irregularly-sampled time series, in: Advances in Neural Information Processing Systems, volume 32, 2019.

[3] P. Kidger, J. Morrill, J. Foster, T. Lyons, Neural controlled differential equations for irregular time series, in: Advances in Neural Information Processing Systems, volume 33, 2020, pp. 6696–6707.

[4] Y. Chen, K. Ren, Y. Wang, Y. Fang, W. Sun, D. Li, Contiformer: Continuous-time transformer for irregular time series modeling, in: Advances in Neural Information Processing Systems, volume 36, 2023.

[5] A. Gu, K. Goel, C. Ré, Efficiently modeling long sequences with structured state spaces, in: International Conference on Learning Representations, 2022. URL: https://openreview.net/forum?id=uYLFoz1vlAC.

[6] A. Gupta, A. Gu, J. Berant, On the parameterization and initialization of diagonal state space models, in: Advances in Neural Information Processing Systems, volume 35, 2022, pp. 35971–35983.

[7] J. T. H. Smith, A. Warrington, S. W. Linderman, Simplified state space layers for sequence modeling, in: International Conference on Learning Representations, 2023. URL: https://openreview.net/forum?id=Ai8Hw3AXqks.

[8] A. Orvieto, S. L. Smith, A. Gu, A. Fernando, C. Gulcehre, R. Pascanu, S. De, Resurrecting recurrent neural networks for long sequences, in: Proceedings of the 40th International Conference on Machine Learning, volume 202 of *Proceedings of Machine Learning Research*, 2023, pp. 26670–26698.

[9] A. Gu, T. Dao, Mamba: Linear-time sequence modeling with selective state spaces, in: Proceedings of the First Conference on Language Modeling, 2024. URL: https://openreview.net/forum?id=tEYskw1VY2.

[10] B. Yu, J. Kim, Wave function backpropagation with explicit temporal-interval dynamics, 2026. URL: https://arxiv.org/abs/2609.00503, the 38th IEEE International Conference on Tools with Artificial Intelligence (ICTAI).

[11] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, D. Duvenaud, Neural ordinary differential equations, in: Advances in Neural Information Processing Systems, volume 31, 2018.

[12] E. De Brouwer, J. Simm, A. Arany, Y. Moreau, GRU-ODE-Bayes: Continuous modeling of sporadically-observed time series, in: Advances in Neural Information Processing Systems, volume 32, 2019.

[13] R. Hasani, M. Lechner, A. Amini, D. Rus, R. Grosu, Liquid time-constant networks, in: Proceedings of the AAAI Conference on Artificial Intelligence, volume 35, 2021, pp. 7657–7666. doi:10.1609/aaai.v35i9.16936.

[14] D. Neil, M. Pfeiffer, S.-C. Liu, Phased LSTM: Accelerating recurrent network training for long or event-based sequences, in: Advances in Neural Information Processing Systems, volume 29, 2016.

[15] M. Schirmer, M. Eltayeb, S. Lessmann, M. Rudolph, Modeling irregular time series with continuous recurrent units, in: Proceedings of the 39th International Conference on Machine Learning, volume 162 of *Proceedings of Machine Learning Research*, 2022, pp. 19388–19405. URL: https://proceedings.mlr.press/v162/schirmer22a.html.

[16] T. K. Rusch, S. Mishra, UnICORNN: A recurrent model for learning very long time dependencies, in: Proceedings of the 38th International Conference on Machine Learning, volume 139 of *Proceedings of Machine Learning Research*, 2021, pp. 9168–9178. URL: https://proceedings.mlr.press/v139/rusch21a.html.

[17] J. Oskarsson, P. Sidén, F. Lindsten, Temporal graph neural networks for irregular data, in: Proceedings of the 26th International Conference on Artificial Intelligence and Statistics, volume 206 of *Proceedings of Machine Learning Research*, PMLR, 2023, pp. 4515–4531. URL: https://proceedings.mlr.press/v206/oskarsson23a.html.

[18] B. Lusch, J. N. Kutz, S. L. Brunton, Deep learning for universal linear embeddings of nonlinear dynamics, Nature Communications 9 (2018) 4950. doi:10.1038/s41467-018-07210-0.

[19] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, in: Advances in Neural Information Processing Systems, volume 30, 2017.

[20] N. Hansen, H. Su, X. Wang, TD-MPC2: Scalable, robust world models for continuous control, in: International Conference on Learning Representations, 2024. URL: https://openreview.net/forum?id=Oxh5CstDJU.

[21] A. Scannell, M. Nakhaeinezhadfard, K. Kujanpää, Y. Zhao, K. S. Luck, A. Solin, J. Pajarinen, Discrete codebook world models for continuous control, in: International Conference on Learning Representations, 2025.

[22] V. Sitzmann, J. N. P. Martel, A. W. Bergman, D. B. Lindell, G. Wetzstein, Implicit neural representations with periodic activation functions, in: Advances in Neural Information Processing Systems, volume 33, 2020, pp. 7462–7473.

[23] E. Todorov, T. Erez, Y. Tassa, MuJoCo: A physics engine for model-based control, in: 2012 IEEE/RSJ International Conference on Intelligent Robots and Systems, 2012, pp. 5026–5033. doi:10.1109/IROS.2012.6386109.

[24] I. Silva, G. Moody, R. Mark, L. A. Celi, Predicting mortality of ICU patients: The PhysioNet/Computing in Cardiology Challenge 2012, 2012. URL: https://physionet.org/content/challenge-2012/1.0.0/, physioNet, version 1.0.0.

[25] T. Pollard, B. E. Moody, L.-w. H. Lehman, B. J. Gow, C. Fernandes, C. Xie, A. Johnson, R. G. Mark, T. Heldt, PhysioNet as a global platform for biomedical research, Nature Health 1 (2026) 792–795. doi:10.1038/s44360-026-00096-z.